

Optimal Financial Regulation with Imperfect Instruments*

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Abstract

This paper characterizes optimal corrective regulation when some decisions cannot be perfectly regulated, offering a unified way to analyze imperfect financial regulation. In a wide range of economies, Pigouvian wedges and leakage elasticities — the responses of imperfectly regulated decisions to policy changes — determine the second-best regulation. Reverse leakages determine the value of relaxing regulatory constraints. The framework jointly addresses seemingly distinct problems in financial regulation, including shadow banking, risk-shifting, and political constraints on intervention. We then apply it to bank regulation with shadow banks, combining direct empirical measurement of leakages with a quantitative model of Pigouvian wedges. The application provides insights into when financial regulation should tighten or loosen, and quantifies optimal policy across several regimes.

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1 Introduction

Many economic policies aim to correct externalities. Financial regulation, a prime example, addresses market failures such as fire sales and distortive government bailouts (e.g., [Lorenzoni, 2008](#); [Bianchi, 2011, 2016](#); [Dávila and Korinek, 2018](#)). In practice, however, financial regulation operates in the face of imperfect instruments and “unintended consequences”, typically in the form of regulatory arbitrage (e.g., [Adrian and Ashcraft, 2016](#); [Hachem, 2018](#)). The resulting second-best problem seems daunting, since many distinct market failures and instrument imperfections are at play.

In this paper, we identify the principles that determine second-best corrective policy, and use them for quantitative welfare analysis. We study a canonical economy with financial frictions.¹ Heterogeneous intermediaries choose investment and financing that may generate externalities. A planner sets corrective taxes or regulations, but the instruments are constrained. For example, the planner may i) be unable to regulate certain agents or decisions, ii) be forced to set uniform regulation across agents or decisions, or iii) face costs that rise with the size of regulation. In financial regulation, these constraints include familiar issues, such as the presence of unregulated shadow banks, risk-shifting, and the political costs of financial regulation.

Our core insight is that all of these imperfections — typically treated as distinct phenomena — obey the same simple economics. We first show that the marginal welfare effect of any policy change depends only on *Pigouvian wedges* and *policy elasticities*. Pigouvian wedges measure the gap between existing corrective regulations and marginal externalities — a positive wedge, for instance, suggests overregulation relative to the Pigouvian “polluter pays” principle. Policy elasticities measure the equilibrium responses of different decisions to changes in regulation. Intuitively, our characterization shows that policy changes that discourage underregulated decisions or encourage overregulated decisions raise welfare. A less intuitive implication is that nothing else additionally matters for optimal policy, even with market incompleteness, financial frictions, and default.

Three results on optimal second-best policy follow from this characterization. First, even *perfectly regulated decisions* — which are not associated with any binding constraint

¹Focusing on financial regulation is natural, since financial activity is inherently hard to regulate ([Arseneau, Brang, Darst, Faber, Rappoport, and Vardoulakis, 2023](#)). However, our results also apply to classical price-theoretic environments with consumption/production decisions, as well as game-theoretic environments, as we illustrate in the Online Appendix.

faced by the planner — are subject to a second-best adjustment. The adjustment is positive, meaning stricter regulation than the Pigouvian level, if imperfectly regulated decisions are underregulated complements or overregulated substitutes; sub-Pigouvian regulation arises for underregulated substitutes or overregulated complements. *Leakage elasticities* measure how imperfectly regulated decisions respond to policy changes and therefore play a key role in determining second-best regulation. These leakages are often quantifiable, allowing us to move beyond an “anything goes” view of second-best regulation.

Second, we characterize the optimal second-best regulation of *imperfectly regulated* decisions. By definition, these regulations are subject to binding constraints, but the planner may nonetheless have degrees of freedom in choosing them — for instance, if a constraint dictates that two decisions must be taxed at a uniform rate, while the level of the uniform tax can be chosen freely. When choosing regulations on imperfectly regulated decisions, the planner must not only consider direct effects, but also more nuanced feedback effects. In general, the optimal second-best regulation depends on leakage and *reverse leakage* elasticities; the latter capture how perfectly regulated decisions adjust to regulating imperfectly regulated ones.

Building on these insights, we derive optimal second-best regulation of imperfectly regulated decisions in two common scenarios. If taxes must be *uniform* across heterogeneous agents or decisions, then the optimal regulation is a weighted average of distortions, where the appropriate weights are augmented to incorporate reverse leakage elasticities. If a subset of regulations is subject to convex costs, then the optimal regulation is given by an attenuated version of the first-best policy. Reverse leakage further attenuates the optimal regulation.

Finally, we characterize the social value of relaxing the constraints faced by a planner who is implementing the optimal second-best policy. This matters for reforms that bring shadow banking under the regulatory perimeter. Whenever perfectly and imperfectly regulated decisions are *either* complements or substitutes, reverse leakage *attenuates* the welfare effect of regulating imperfectly regulated decisions — a result that connects to the Le Chatelier principle. As a result, regulatory reform may be desirable, but the quantitative case in favor of reform is weaker because general equilibrium adjustment offsets part of the direct gain.

To translate our results into quantitative insights for imperfect financial regulation, we consider an environment in which both regulated banks and unregulated shadow banks

make leverage and investment decisions. Practical regulation in this setting is imperfect in several ways. Shadow banks are generally thought to contribute to systemic risk, but remain mostly unregulated. Moreover, common financial regulations, such as the Basel Accords, concentrate on ratios of assets to liabilities, so that the scale of banks' operations is unregulated. Therefore, this application gives rise to a rich quantitative mix of imperfections, which our theoretical work helps us to disentangle. We conduct two complementary exercises.

First, without fully specifying the model, we use empirical leakage elasticities to quantify the welfare effects of imperfect bank regulation as a function of Pigouvian wedges. Estimates in [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) imply that a one-percentage-point increase in traditional banks' leverage regulation, for example via standard capital requirements, reduces traditional bank lending by 5%, while increasing shadow bank lending by 4%. When combined with our theoretical results, these measures imply that marginal increases in leverage regulation are welfare-increasing despite the presence of shadow banks, unless i) traditional banks are already severely overregulated, or ii) shadow banks impose quantitatively larger externalities than traditional banks' investment.

Next, we build a full general equilibrium model to quantify externalities and compute optimal second-best regulation. The model includes three distortions that are central to financial regulation in practice: bailouts, pecuniary/fire-sale externalities, and distorted beliefs. We use measured leakage elasticities as key calibration targets. We study four policy scenarios: First, we solve for the first-best benchmark, which sets all Pigouvian wedges to zero.

Second, we consider a situation where the planner can regulate traditional banks' and shadow banks' leverage, but investment scale remains a free choice. We find that the second-best policy overregulates traditional banks but underregulates shadow banks relative to the first-best. Our theory provides an intuitive interpretation: The scales of investment in both traditional and shadow banks are substitutes, calling for underregulation, while leverage and scale within an institution are complements, calling for overregulation. Given the Pigouvian wedges implied by our calibration, the latter effect dominates for traditional banks, and the former for shadow banks.

Third, we consider a situation where the planner can regulate only traditional banks' leverage — which is closest to current regulatory practice. This case has three imperfections and hence three relevant leakage elasticities. Given our calibration, the strong substitutability between traditional banks' leverage and shadow banks' investment

dominates, justifying the optimal underregulation of traditional banks.

Finally, to evaluate proposals that subject shadow banks to the same regulation as traditional banks, we illustrate the form of the optimal uniform regulation of leverage, and consider the social value of marginally relaxing the regulatory constraint on shadow bank leverage.

Overall, our quantitative results highlight that leakage elasticities featuring both substitutability and complementarity naturally emerge and nontrivially interact in common regulatory scenarios. A quantitative analysis, which incorporates different banks' different decisions, is typically necessary to determine optimal second-best policy. A useful contribution of our theoretical results is to provide a clear and interpretable quantitative analysis of the associated welfare implications, despite the complexity of the relevant effects.

Related Literature. A growing literature studies financial regulation with imperfections. One key imperfection is that financial intermediaries are heterogeneous (e.g., in size) but typically subject to uniform regulation (Corbae and D'Erasmus, 2021). Unregulated shadow banks have also been highlighted as a source of imperfect regulation. Theoretical work notes the unintended consequences of liquidity policies (Hachem and Song, 2021; Grochulski and Zhang, 2019; Hachem and Kuncel, 2025) and capital requirements (Plantin, 2015; Huang, 2018; Martinez-Miera and Repullo, 2019; Farhi and Tirole, 2021; Clayton and Schaab, 2021), as well as the general costs and benefits of shadow banks (e.g., Gennaioli, Shleifer, and Vishny, 2013; Moreira and Savov, 2017; Ordoñez, 2018). Another strand of work focuses on imperfect regulation of financiers' portfolio choices, with foundational work emphasizing risk-shifting (Kim and Santomero, 1988; Allen and Gale, 2000; Hellmann, Murdock, and Stiglitz, 2000), and more recent research emphasizing carbon emissions (Oehmke and Opp, 2022). Other imperfections arise from conflicts between financial regulators, as well as technological innovations (Korinek, 2017; Korinek, Montecino, and Stiglitz, 2022). We emphasize the common economics behind second-best regulation in these seemingly distinct scenarios.

Quantitative models (Bengui and Bianchi, 2022; Begenau and Landvoigt, 2022; Dempsey, 2025) provide general equilibrium assessments of bank leverage regulation with shadow banks. Empirically, Acharya, Schnabl, and Suarez (2013), Demyanyk and Loutskina (2016), and Buchak, Matvos, Piskorski, and Seru (2018, 2024b), among others, measure the associated leakages. We show how measured leakages can be used directly

for quantitative policy statements. Unlike existing work, our quantitative application includes a variety of imperfections and market failures — and our theoretical results contribute crucial tools for economic interpretation of this complex setting.

Our focus on imperfect regulation relates to canonical models of second-best corrective taxes (e.g., [Salanié, 2011](#)). None of this literature considers economies with incomplete markets or financial frictions. Our results also provide a novel narrative for classical results: For instance, the optimal tax formulae in [Diamond \(1973\)](#) are seemingly distinct from the characterization of second-best policy in [Lipsey and Lancaster \(1956\)](#) or the [Tinbergen \(1952\)](#) Rule, but these results can all be derived as corollaries of our general results. Moreover, second-best corrective regulation is often discussed in the context of environmental policy and congestion (e.g., [Bovenberg and Goulder, 2002](#)), as well as rent-seeking (e.g., [Rothschild and Scheuer, 2016](#)).

2 General Framework

This section presents our general framework. In [Section 3](#), we use it to derive the principles governing optimal regulation with imperfect instruments. In [Section 4](#), we then show how these principles can be used to quantify optimal constrained policy in an application with pervasive imperfections.

2.1 Environment

We consider an economy with two dates, $t \in \{0, 1\}$, and a single consumption good. At date 1, there is a continuum of possible states $s \in S$, with a cumulative distribution function $F(s)$. There are two groups of agents: investors and creditors.

Investors. There is a finite number of investor types indexed by $i, j \in \mathcal{I} = \{1, 2, \dots, I\}$. In the application below, for instance, we will have $I = 2$ types: traditional banks and unregulated shadow banks.

Each investor type makes $N \geq 0$ investment/financing decisions, which we collect in a vector $\mathbf{x}^i \in \mathbb{R}^N$. We use this abstract notation for financing/investment decisions, but it maps directly into applied settings.² Concretely, collect investors' liabilities — such as promised repayments on debt, bonds, or deposits, or the amount of equity sold to outsiders

²Section [A](#) of the Online Appendix includes explicit definitions of all vectors and matrices used.

— in a vector of financing decisions $\mathbf{b}^i$. Assets — such as a portfolio of securities, lending opportunities, or projects — can be collected in a vector of investment decisions $\mathbf{k}^i$. The overall vector of decisions is then simply $\mathbf{x}^i = (\mathbf{b}^i, \mathbf{k}^i)$, measuring both sides of investors' balance sheets.

Investor i 's preferences are represented by

$$u^i \left(c_0^i, \{c_1^i(s)\}_{s \in S}, \bar{\mathbf{x}} \right), \quad (1)$$

where c_0^i and $c_1^i(s)$ denote investor i 's consumption at date 0 and date 1 in state s , while $\bar{\mathbf{x}} = \{\bar{\mathbf{x}}^j\}_{j \in \mathcal{I}}$ denotes the $N \cdot I$ decisions made by all investor types in equilibrium. Since each individual investor is atomistic, her decisions $\mathbf{x}^i$ do not affect the aggregate decisions $\bar{\mathbf{x}}^i$ of her own type, nor the aggregate decisions $\bar{\mathbf{x}}^j$ of other types. The dependence of utility on $\bar{\mathbf{x}}$ captures externalities across agents. In financial regulation, this covers bailout externalities, fire-sale pecuniary externalities, and “internalities” from distorted beliefs, all of which we consider in Section 4.

Investor i faces the following budget constraints:

$$c_0^i = n_0^i + Q^i(\mathbf{x}^i) - \Upsilon^i(\mathbf{x}^i) - \boldsymbol{\tau}^i \cdot \mathbf{x}^i + T^i \quad (2)$$

$$c_1^i(s) = n_1^i(s) + \rho^i(\mathbf{x}^i, s), \quad \forall s, \quad (3)$$

where n_0^i and $n_1^i(s)$ denote investor endowments. At date 0, investor i raises financing according to the pricing schedule $Q^i(\mathbf{x}^i)$, which is determined endogenously in equilibrium. Investor i 's decisions also induce a cost $\Upsilon^i(\mathbf{x}^i) \geq 0$. This term can capture technological adjustment costs for investment decisions $\mathbf{k}^i$, or financing frictions for financing decisions $\mathbf{b}^i$. The term $\boldsymbol{\tau}^i \cdot \mathbf{x}^i$ represents corrective taxes, subsidies, or regulations, where $\boldsymbol{\tau}^i \in \mathbb{R}^N$, and $T^i \in \mathbb{R}$ denotes the lump-sum transfer or tax that investor i receives. At date 1, the investor enjoys a return $\rho^i(\mathbf{x}^i, s)$ from her investments. This function can be interpreted as the investor's residual claim on her investments, after paying back all liabilities to creditors.

Creditors. A unit mass of creditors has preferences

$$u^C \left(c_0^C, \{c_1^C(s)\}_{s \in S}, \bar{\mathbf{x}} \right), \quad (4)$$

where c_0^C and $c_1^C(s)$ denote creditors' consumption at date 0 and at date 1 in state s . As above, $\bar{\mathbf{x}}$ denotes the collection of decisions made by all investors. Creditors can be interpreted as a type of investor who is only allowed to fund other investors.³

Creditors face the following budget constraints:

$$c_0^C = n_0^C - \sum_i h_i^C Q^i(\bar{\mathbf{x}}^i) \quad (5)$$

$$c_1^C(s) = n_1^C(s) + \sum_i h_i^C \rho_i^C(\bar{\mathbf{x}}^i, s), \quad \forall s, \quad (6)$$

where n_0^C and $n_1^C(s)$ denote creditors' endowments. At date 0, creditors choose to fund a share h_i^C of each investor i 's financing needs $Q^i(\cdot)$. At date 1, creditors receive repayments $\rho_i^C(\bar{\mathbf{x}}^i, s)$ from investor i . Notice that the general formulation of investor returns $\rho^i(\cdot)$ and creditor repayments $\rho_i^C(\cdot)$ can accommodate deadweight losses associated with investors' default. By suitably interpreting creditors' preferences, our model can capture nonpecuniary benefits/convenience yields from particular forms of financing, as in, e.g., [Stein \(2012\)](#) or [Sunderam \(2015\)](#).

Equilibrium. For given corrective taxes/subsidies/regulations $\{\tau^i\}_i$ and lump-sum transfers $\{T^i\}_i$, an *equilibrium* consists of consumption allocations $\{c_0^i, c_1^i(s)\}_i$ and $\{c_0^C, c_1^C(s)\}$, investors' decisions $\{\bar{\mathbf{x}}^i\}_i$, creditors' funding decisions $\{h_i^C\}_i$, financing schedules $\{Q^i(\cdot)\}_i$, investors' net investment returns $\{\rho^i(\cdot)\}_i$, and creditors' repayments $\{\rho_i^C(\cdot)\}_i$ such that i) investors maximize utility, (1), subject to budget constraints (2) and (3); ii) creditors maximize utility, (4), subject to budget constraints (5) and (6); iii) the planner's budget is balanced, so that $\sum_i T^i = \sum_i \tau^i \cdot \mathbf{x}^i$; iv) investors' decisions are consistent in the aggregate, that is, $\mathbf{x}^i = \bar{\mathbf{x}}^i$, $\forall i$; v) financing decisions satisfy market clearing, that is, $h_i^C = 1$, $\forall i$.

Note that individual investors are price-takers in the sense that they take the pricing schedule $Q^i(\cdot)$ as given. Nonetheless, each individual internalizes that her investment/financing decisions $\mathbf{x}^i$ affect the amount of financing $Q^i(\mathbf{x}^i)$ that she can raise. This equilibrium notion is standard in models that allow for default (e.g., [Dubey, Geanakoplos, and Shubik, 2005](#)). We proceed as if the model is well-behaved, discussing

³In earlier versions of this paper, we allowed for investors to invest in each other's liabilities and for creditors' decisions to also be associated with welfare-relevant externalities. Since the main insights are identical in both formulations, we adopt the current formulation, which substantially simplifies the notation.

the necessary regularity conditions in each application. Assuming $T^i = \boldsymbol{\tau}^i \cdot \mathbf{x}^i$, $\forall i$ — instead of the less restrictive $\sum_i T^i = \sum_i \boldsymbol{\tau}^i \cdot \mathbf{x}^i$ — makes Pigouvian taxes equivalent to quantity regulation.

2.2 Imperfect Policy Instruments

The planner in this model has $N \cdot I$ corrective policy instruments, one for each investor decision, collected in $\boldsymbol{\tau} = \{\boldsymbol{\tau}^i\}_{i \in \mathcal{I}}$. We formalize *imperfect instruments* by assuming that the planner chooses regulations subject to a vector-valued constraint

$$\Phi(\boldsymbol{\tau}) \leq 0. \quad (7)$$

The function $\Phi : \mathbb{R}^{IN} \rightarrow \mathbb{R}^M$, where M is the number of constraints, defines feasible regulations.

Example (Common Cases in Financial Regulation). Our main results — Propositions 1 and 2 — are valid for a general $\Phi(\cdot)$. Several familiar imperfections are nested in this formalism:

1. *Unregulated investors:* The function $\Phi(\cdot)$ can mandate that $\boldsymbol{\tau}^i \equiv 0$ for an “unregulated” type of investor.⁴ Unregulated shadow banks are a widely modeled example of this type of imperfection.
2. *Unregulated decisions:* The function $\Phi(\cdot)$ can mandate that $\tau_n^i \equiv 0$ for all elements n of the vector $\boldsymbol{\tau}^i$ that correspond to an unregulated decision *within* a type of investor, even if that type is not entirely unregulated. A frequently discussed example in financial regulation is risk-shifting (or asset substitution): within a bucket of investments that regulators treat as equivalent, financial intermediaries can choose the share that is allocated to riskier projects, and this decision is effectively unregulated. Another case study is financial regulation in the presence of environmental externalities, or climate finance. Finally, financial regulations typically constrain ratios (e.g., of risk-weighted assets to liabilities), leaving the overall scale of the balance sheet as a free — and hence unregulated — decision.
3. *Uniformity:* One can specify a linear function $\Phi(\cdot)$ that imposes uniform regulation on potentially heterogeneous investors. This is common in financial

⁴Formally, one would stack two constraints: $\boldsymbol{\tau}^i \leq 0$ and $\boldsymbol{\tau}^i \geq 0$, so that $M = 2 \cdot N$.

regulation: capital and liquidity requirements generally do not depend on observable institutional features.

4. *Costs of regulation/political economy:* Appealing to the duality between constraints and costs, we can also interpret $\Phi(\boldsymbol{\tau})$ as defining the cost of setting regulations. A common observation is that industry lobbying and other political constraints make tougher financial regulation costly. Formally, a quadratic constraint $\Phi(\boldsymbol{\tau}) \equiv \frac{1}{2}\boldsymbol{\tau}'\mathbf{B}\boldsymbol{\tau}$ for a given matrix $\mathbf{B}$, can represent convex costs of regulation to capture such phenomena. In addition, if the costs of regulation induce sparsity (Tibshirani, 1996; Gabaix, 2014) — for instance, when based on the L^1 norm of $\boldsymbol{\tau}$ — the set of unregulated agents or commodities arises endogenously.

Most of these examples are associated with relatively simple (i.e., linear or quadratic) constraints $\Phi(\boldsymbol{\tau})$. One could also imagine more complex scenarios, such as dynamic regulatory arbitrage or financial innovation, that would give rise to nonlinear constraints. Our theoretical results apply to these settings as long as the implied constraint function is well-behaved.

In the next section, we characterize second-best optimal regulation in this general class of economies. These results are applied to study financial regulation with both unregulated investors (shadow banks) and unregulated decisions (balance sheet scale) in Section 4, where we also explore optimal uniform regulations. In a previous working paper — available on request — we demonstrated further applications of our framework, including financial regulation in the presence of environmental externalities. However, we note that our general results do not depend on the financing/investment model studied here, and would also apply to price- or game-theoretic environments. We illustrate this possibility in Section C of the Online Appendix.

3 Optimal Corrective Policy with Imperfect Instruments

We now study the problem of a planner who optimally chooses imperfect corrective regulation. We abstract from redistributational considerations, and focus on the corrective nature of the regulation. Therefore, we assess the aggregate gains/losses of a marginal

policy change by aggregating money-metric welfare changes across agents.⁵ That is, the planner evaluates the desirability of a marginal change in a given variable (or vector) z , denoted by $\frac{dW}{dz}$, according to

$$\frac{dW}{dz} \equiv \sum_{i \in \mathcal{I}} \frac{\frac{dV^i}{dz}}{\lambda_0^i} + \frac{\frac{dV^C}{dz}}{\lambda_0^C},$$

where $\frac{dV^i}{dz}$ ($\frac{dV^C}{dz}$) denotes the change in investor i 's (creditor's) indirect utility in equilibrium and $\lambda_0^i > 0$ ($\lambda_0^C > 0$) denotes investor i 's (creditor's) date 0 marginal value of consumption.

To characterize the marginal welfare effect of adjusting regulations, it is useful to first define the *marginal distortions/externalities* associated with the vector $\bar{\mathbf{x}}$ of decisions:

$$\boldsymbol{\delta} = - \left(\sum_{j \in \mathcal{I}} \frac{1}{\lambda_0^j} \frac{\partial u^j}{\partial \bar{\mathbf{x}}} + \frac{1}{\lambda_0^C} \frac{\partial u^C}{\partial \bar{\mathbf{x}}} \right). \quad (\text{Marginal Distortion})$$

The vector $\boldsymbol{\delta}$ contains one distortion for each investor decision (of which there are $N \cdot I$ in the economy). We define the distortion as the negative of marginal utility, so that $\boldsymbol{\delta}$ stands for “damage” or negative externalities. In addition, we define *Pigouvian wedges* as the difference between regulations and marginal distortions:

$$\boldsymbol{\omega} = \boldsymbol{\tau} - \boldsymbol{\delta}. \quad (\text{Pigouvian Wedge})$$

The well-known “polluter pays” principle of corrective regulation (Pigou, 1920) suggests that all wedges should be set to zero. The first-best Pigouvian tax is given by $\boldsymbol{\tau}^* = \boldsymbol{\delta}$.⁶

We focus instead on situations where regulatory imperfections prevent the first-best allocation. In these settings, positive Pigouvian wedges represent overregulation, and negative ones underregulation, relative to the Pigou principle. Of course, our notation also allows for some decisions to be underregulated while others are overregulated. The vector $\boldsymbol{\omega}$ simply collects wedges for all $N \cdot I$ decisions made by investors, and different

⁵This approach is akin to maximizing Kaldor-Hicks efficiency — see Dávila and Schaab (2025) — or to imposing equal generalized social marginal welfare weights, in the sense of Saez and Stantcheva (2016). Section C.3 of the Online Appendix explains how to consider traditional social welfare functions and redistributive considerations.

⁶Since our model features incomplete markets and default, the first-best regulation is interpreted as the one that achieves constrained efficiency. Beyond this benchmark, a planner could potentially raise welfare further if she were able to implement better risk sharing, or lower incidence of costly default, than is achievable with the financing instruments that investors use. Our focus on constrained efficiency is shared with the majority of the literature on regulation with financial frictions.

elements can have different signs.

Lemma 1 highlights the role of Pigouvian wedges and policy elasticities when evaluating arbitrary policy changes:

Lemma 1. (*Marginal Welfare Effects of Regulation*) *The marginal welfare effects of varying regulations τ , $\frac{dW}{d\tau}$, are given by*

$$\frac{dW}{d\tau} = \frac{d\mathbf{x}}{d\tau} (\tau - \delta) = \frac{d\mathbf{x}}{d\tau} \boldsymbol{\omega}, \quad (8)$$

where $\frac{d\mathbf{x}}{d\tau}$ is the Jacobian matrix of policy elasticities, of dimension $NI \times NI$.

Lemma 1 shows that the welfare impact of regulation changes depends only on policy elasticities and Pigouvian wedges. The matrix $\frac{d\mathbf{x}}{d\tau}$ of *policy elasticities* — borrowing the terminology of Hendren (2016) — captures the full equilibrium response that a particular change in regulation has on all agents' decisions. The vector $\boldsymbol{\omega}$ of Pigouvian wedges captures the extent to which an agent's decision is underregulated or overregulated. The overall welfare effect of varying a particular regulation corresponds to the sum of products of the relevant policy elasticities and Pigouvian wedges, where regulations that i) decrease underregulated decisions (with a negative wedge) or ii) increase overregulated decisions (with a positive wedge) in equilibrium are welfare-improving.

3.1 Second-Best Policy: Perfectly Regulated Decisions

To characterize the optimal second-best policy, we let $\mathbf{x}^R$ denote all *perfectly regulated* decisions made by investors. These are decisions for which the planner can choose the corresponding tax or regulation without a binding constraint. We let $\mathbf{x}^U$ denote the remaining, *imperfectly regulated* decisions.⁷ We use the same partition for all objects in the model: for instance, τ^R denotes the tax/regulation on perfectly regulated decisions, and $\boldsymbol{\omega}^U$ the Pigouvian wedge on imperfectly regulated decisions. We also apply it to derivatives: For instance, $\frac{d\mathbf{x}^U}{d\tau^R}$ is the matrix of *leakage elasticities*, which capture how imperfectly regulated decisions respond to changes in the regulations that can be freely adjusted.⁸

⁷Formally, let the vector $\boldsymbol{\mu} \in \mathbb{R}^M$ denote the vector of Lagrange multipliers associated with the regulatory constraints in Equation (7). If an element of $\frac{d\boldsymbol{\Phi}}{d\tau} \boldsymbol{\mu}$ is zero (not zero), then the corresponding element of decisions $\mathbf{x}$ is a perfectly (imperfectly) regulated decision.

⁸All matrices are defined more rigorously in the appendix. When there are R perfectly regulated decisions and U imperfectly regulated ones — where $R + U = N \cdot I$ — then the leakage matrix $\frac{d\mathbf{x}^U}{d\tau^R}$ has

Proposition 1 shows how leakage elasticities determine the second-best policy.

Proposition 1. *(Second-Best Policy: Perfectly Regulated Decisions) The optimal second-best regulation of perfectly regulated decisions satisfies*

$$\boldsymbol{\tau}^R = \boldsymbol{\delta}^R + \left(-\frac{d\mathbf{x}^R}{d\boldsymbol{\tau}^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^R} \boldsymbol{\omega}^U, \quad (9)$$

where $\boldsymbol{\delta}^R$ is a vector of marginal distortions, $\boldsymbol{\omega}^U = \boldsymbol{\tau}^U - \boldsymbol{\delta}^U$ is a vector of Pigouvian wedges, $\frac{d\mathbf{x}^R}{d\boldsymbol{\tau}^R}$ is a matrix of own perfectly regulated policy elasticities, and $\frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^R}$ is a matrix of leakage elasticities.

Proposition 1 shows that optimal regulation has two parts: the marginal distortion $\boldsymbol{\delta}^R$ (following the Pigou principle), and a second-best correction driven by leakage elasticities and Pigouvian wedges. A useful heuristic for interpreting this expression is that $\left(-\frac{d\mathbf{x}^R}{d\boldsymbol{\tau}^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^R} \equiv -\frac{d\mathbf{x}^U}{d\mathbf{x}^R}$ is positive (negative) if regulated and unregulated decisions are gross substitutes (complements).

To provide more intuition, it is useful to go back to the examples from financial regulation that we laid out in Section 2. Take the first example given above — unregulated investors/shadow banks. As we show more formally in our application below, unregulated shadow bank lending typically increases when bank regulations are tightened. Bank and shadow bank decisions are substitutes, so the leakages $\frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^R}$ are positive. If shadow banks impose externalities but face no corrective policy, their Pigouvian wedges $\boldsymbol{\omega}^U$ are negative. Equation (9) then yields $\boldsymbol{\tau}^R < \boldsymbol{\delta}^R$, since the second-best correction is negative overall. More generally, whenever imperfectly regulated decisions are i) underregulated and ii) substitutes to regulated ones, the second-best regulation $\boldsymbol{\tau}^R$ inherits underregulation relative to the Pigouvian benchmark.

For another relevant case, consider the second example — unregulated decisions due to risk-shifting. Some of the key tools of financial regulation are constraints/taxes on leverage, while agency theories in finance and banking (e.g., Jensen and Meckling, 1976; Allen and Gale, 2000) suggest that levered institutions are more likely to engage in high-risk investment, so that leverage and risk-taking are complements. Therefore, if risk-taking is imperfectly regulated and imposes negative externalities, Equation (9) yields $\boldsymbol{\tau}^R > \boldsymbol{\delta}^R$.⁹ More generally, whenever imperfectly regulated decisions are i)

dimension $R \times U$.

⁹Our application below shows that the same qualitative result applies when the scale of banks' balance

underregulated and ii) complements to regulated ones, the second-best regulation τ^R switches to overregulation relative to Pigou.

What emerges from this discussion is that there are forces pushing for both overregulation and underregulation in the second-best scenario, and that both effects are present in standard problems of financial regulation. Hence, the picture of second-best regulation is nuanced. Based on the cases above, for instance, one cannot conclude whether one should optimally over- or underregulate bank leverage in the presence of shadow banks. However, it would also be too pessimistic to conclude that “anything goes”. Indeed, our results provide a clear guide to interpreting the economic forces at play: underregulated substitutes (and, conversely, overregulated complements) generate optimal regulations below the Pigouvian benchmark; while underregulated complements (and, conversely, overregulated substitutes) generate optimal regulations above the Pigouvian benchmark. The nature of the second-best policy then becomes the empirical question of measuring the relevant leakages and wedges. Our application in the next section demonstrates several approaches to this problem.

It is important to note that distortions δ^R and δ^U are, in general, endogenous to the regulatory regime. For instance, if a regulator underregulates banks’ leverage in order to prevent leakage to shadow banks (substitutes), it could still be the case that the marginal social distortion δ^R grows as a result of underregulation. In our quantitative application below, we solve for endogenous distortions for common market failures in financial regulation — e.g., fire sales, bank bailouts, and distorted beliefs — and take this mechanism into account.

An interesting alternative interpretation of Equation (9) is that a second-best planner uses the R free instruments τ^R (on the left-hand side) to target the $R + U$ distortions contained in δ^R and ω^U (on the right-hand side). Only when $\omega^U = 0$ does a first-best outcome obtain, consistent with the [Tinbergen \(1952\)](#) rule. More generally, Equation (9) refines the logic: with insufficient policy instruments, the optimal regulation tries to set a weighted sum of all wedges to zero, with weights shaped by leakage elasticities.

Proposition 1 is still incomplete. To see this, consider uniform regulation. If the same regulations have to be applied to a number of heterogeneous investor types, then all decisions of those types would be classified as imperfectly regulated by our terminology, because they are associated with a binding constraint faced by the regulator. However,

sheets, as opposed to risk, is an unregulated decision.

the regulator still has degrees of freedom, namely the level of the uniform tax or regulation τ^U . Similarly, in the fourth example — where policy is costly due to political economy — all regulations are imperfectly regulated, but the regulator still needs to choose the level of the costly tax/regulation τ^U . Proposition 1 is silent on these choices, to which we turn in the next section.

3.2 Second-Best Policy: Imperfectly Regulated Decisions

Proposition 2 characterizes the second-best regulation of imperfectly regulated decisions:

Proposition 2. (*Second-Best Policy: Imperfectly Regulated Decisions*) *The optimal second-best regulation of imperfectly regulated decisions satisfies $\frac{dW}{d\tau^U} = \frac{d\Phi}{d\tau^U} \boldsymbol{\mu}$, where $\boldsymbol{\mu}$ is the vector of Lagrange multipliers associated with the constraints on policy instruments, and*

$$\frac{dW}{d\tau^U} = \frac{d\mathbf{x}^U}{d\tau^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\omega}^U, \quad (10)$$

where $\mathbf{I}$ denotes an identity matrix, $\frac{d\mathbf{x}^U}{d\tau^U}$ is a matrix of own imperfectly regulated policy elasticities, $\boldsymbol{\omega}^U$ is the vector of Pigouvian wedges associated with imperfectly regulated decisions, and where we define the matrix $\mathbf{L} = \left(\frac{d\mathbf{x}^U}{d\tau^U}\right)^{-1} \frac{d\mathbf{x}^R}{d\tau^U} \left(\frac{d\mathbf{x}^R}{d\tau^R}\right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R}$.

Equation (10) contains the key economics in this proposition, and shows the marginal welfare effect of tightening the regulation τ^U on imperfectly regulated decisions. We assess this effect while setting the regulation τ^R to its second-best optimal value. To interpret it, consider the case where there are *only* unregulated decisions. Then, by Lemma 1, the welfare effect of raising τ^U would be simply $\frac{d\mathbf{x}^U}{d\tau^U} \boldsymbol{\omega}^U$ — the product of policy elasticities and Pigouvian wedges. Equation (10) suggests that this welfare effect is attenuated by the matrix $\mathbf{L}$. Heuristically, one can write $\mathbf{L} \equiv \frac{d\mathbf{x}^R}{d\tau^U} \frac{d\mathbf{x}^U}{d\mathbf{x}^R}$. This is the product of leakages $\frac{d\mathbf{x}^U}{d\mathbf{x}^R}$, which determine the optimal regulations given in Proposition 1, and *reverse leakages* $\frac{d\mathbf{x}^R}{d\mathbf{x}^U}$, which are novel to this part of the analysis. Proposition 2 does not yield a closed-form expression for the second-best value of τ^U in general. However, it can be used to support two novel economic insights.

First, Equation (10) provides the marginal welfare gain of relaxing constraints on regulation for a planner who is implementing the optimal second-best policy. A novel insight is that reverse leakage tends to attenuate the welfare benefit of relaxing constraints, because the matrix $\mathbf{L}$ is (heuristically) positive whenever perfectly and imperfectly regulated decisions are either complements or substitutes. The economic intuition is

as follows: If perfectly and imperfectly regulated decisions are substitutes, tightening the regulation on imperfectly regulated decisions increases perfectly regulated decisions through reverse leakage. But Proposition 1 shows that perfectly regulated decisions are underregulated at the second-best, with $\omega^R < 0$, so this increase reduces welfare. Conversely, if perfectly and imperfectly regulated decisions are complements, tightening the regulation on imperfectly regulated decisions reduces perfectly regulated decisions through reverse leakage. But Proposition 1 shows that perfectly regulated decisions are overregulated at the second-best, with $\omega^R > 0$, so this decrease also reduces welfare. Intuitively, optimally regulating perfectly regulated decisions reduces the gains from further regulation.

We label $\mathbf{L}$ the attenuation matrix because this result is reminiscent of the Le Chatelier principle in classical economics. As noted by Milgrom (2006), the Le Chatelier principle more generally explains how the direct effect of a parameter change is augmented by feedback in a system. While existing versions of the principle point towards *amplification* by feedback in a system, we find the opposite implication, namely *attenuation*, in terms of welfare effects.

Second, Proposition 2 gives a practical recipe for deriving second-best policies once the constraint function $\Phi(\cdot)$ is specified. We focus on two cases from the examples above: uniform regulation and convex costs.

Practical Scenario: Uniform Regulation. Suppose that all elements of τ^U have to be equal to a uniform regulation, which we denote by $\bar{\tau}^U$, while all other regulations τ^R can be chosen freely. For a concrete example, which we explore in detail below, the elements of τ^U could be leverage regulations on heterogeneous financial institutions. In this case, Proposition 2 implies that the optimal second-best regulation of imperfectly regulated decisions specializes to

$$\bar{\tau}^U = \frac{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\tau^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\delta}^U}{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\tau^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\iota}}, \quad (11)$$

where $\boldsymbol{\iota}$ denotes a vector of ones. Equation (11) shows that the optimal second-best uniform regulation $\bar{\tau}^U$ over a subset of decisions is simply a weighted sum/average of their marginal distortions $\boldsymbol{\delta}^U$. More responsive decisions carry higher weights. As expected, when marginal distortions are symmetric (so $\boldsymbol{\delta}^U = \boldsymbol{\iota} \bar{\delta}$), Equation (11) implies that the first-best regulation $\bar{\tau}^U = \bar{\delta}$ is optimal. However, when marginal distortions are

heterogeneous, the first-best cannot be achieved with uniform regulation. Once again, all regulations are adjusted for the matrix $\mathbf{L}$ that measures the (Le Chatelier) attenuation of welfare effects in general equilibrium.¹⁰

Practical Scenario: Convex Costs of Regulation. Another common imperfection is costly regulation. Formally, we consider a planner who faces quadratic costs of regulation over a subset of decisions, perhaps driven by the political economy barriers to financial regulation, with $\Phi(\boldsymbol{\tau}) = \frac{1}{2}\boldsymbol{\tau}^{U'}\mathbf{B}\boldsymbol{\tau}^U$, for some positive definite matrix $\mathbf{B}$. In this case, Proposition 2 implies that the optimal second-best regulation of imperfectly regulated decisions specializes to

$$\boldsymbol{\tau}^U = (\mathbf{B} + \mathbf{K})^{-1} \mathbf{K} \boldsymbol{\delta}^U, \quad (12)$$

where $\mathbf{K} = \left(-\frac{dx^U}{d\boldsymbol{\tau}^U}\right)(\mathbf{I} - \mathbf{L})$. Equation (12) shows that the optimal policy with quadratic adjustment costs is an attenuated version of the first-best policy. As expected, as costs vanish and $\mathbf{B} \rightarrow 0$, the optimal policy approaches the first-best, so $\boldsymbol{\tau}^U \rightarrow \boldsymbol{\delta}^U$. But as costs grow relative to $\mathbf{K}$, the optimal policy approaches zero, so $\boldsymbol{\tau}^U \rightarrow 0$. The reverse leakage effects discussed above, which make $\mathbf{L}$ positive and $\mathbf{K}$ smaller, further attenuate the optimal choice of $\boldsymbol{\tau}^U$.

In summary, the theory clarifies second-best corrective regulation by characterizing the optimal regulation of perfectly regulated decisions (Proposition 1) and imperfectly regulated decisions (Proposition 2), for a general class of regulatory imperfections. These results support practical analysis and unify classical insights often viewed as distinct. We now apply them to financial regulation, where instruments are notoriously imperfect.

4 Quantitative Application: Imperfect Financial Regulation

Our application in this section provides a roadmap for translating our theoretical results into practical, quantitative insights. We consider an economy with two investor types $i \in \{R, U\}$, where R denotes regulated traditional banks and U denotes unregulated shadow banks. Banks and shadow banks make decisions $\mathbf{x}^i = \{k_0^i, b^i\}$, where k_0^i is initial

¹⁰The insight that uniform regulation of heterogeneous externalities is characterized by a weighted sum/average of distortions can be traced back to Diamond (1973). Indeed, when *all* decisions are uniformly regulated, Equation (11) corresponds to Diamond's result. Equation (11) generalizes his results by allowing for a subset of decisions to be perfectly regulated.

capital invested, such as loans to firms or households, and b^i measures banks' debt/asset ratio or leverage, which expresses the face value of bank borrowing per unit of investment.

Most of our analysis is related to the welfare effects of marginal policy changes $d\boldsymbol{\tau}$, where $\boldsymbol{\tau}$ is the vector of taxes we specify below. These effects will take the same shape as in Lemma 1:

$$\frac{dW}{d\boldsymbol{\tau}} = \sum_{i \in \{R, U\}} \left(\omega_k^i \frac{dk_0^i}{d\boldsymbol{\tau}} + \omega_b^i \frac{db^i}{d\boldsymbol{\tau}} k_0^i \right). \quad (13)$$

It is important to comment briefly on our notation for regulations: we find it convenient to normalize the leverage wedge ω_b^i in this expression so that it is proportional to k_0^i . Therefore, we are also normalizing the associated taxes/regulations. The total corrective tax for a bank is $(\tau_b^i b^i + \hat{\tau}_k^i) k_0^i$, where τ_b^i is the tax on leverage per unit of investment, and $\hat{\tau}_k^i$ measures any additional corrective regulation of investments. This normalization implies that, for a given b^i , the total marginal investment tax is effectively given by $\tau_k^i = \tau_b^i b^i + \hat{\tau}_k^i$. This parametrization of taxes provides a convenient map to practical financial regulations that are based on quantities.

In Section 4.1, we ask how much can be learned from direct measurement of leakage elasticities alone, without fully specifying the remaining model details. Next, in Section 4.2, we specify a full quantitative equilibrium model that delivers interpretable optimal policies across a variety of practical scenarios. For instance, a standard capital requirement on banks can be expressed as $k_0^R (1 - b^R) \geq k_0^R \theta^R$ in this model, where the left-hand side measures bank equity/capital and, on the right-hand side, θ^R measures the required ratio of equity to assets.¹¹ Since k_0^R cancels out from this constraint, the equivalent regulation in our framework is one that sets $\tau_b^i \geq 0$ — since a binding constraint may be associated with the leverage decision b^R — but maintains $\tau_k^i = 0$, so that the scale of bank investments is an unregulated decision.

4.1 Direct Measurement

In this subsection, we focus on the welfare effects of a marginal change in bank capital requirements, which are equivalent to a corrective tax/regulation τ_b^R on traditional banks' leverage decisions. We examine the effects of a marginal increase $d\tau_b^R$, where $\omega_k^i = \tau_k^i - \delta_k^i$ and $\omega_b^i = \tau_b^i - \delta_b^i$ stand for the Pigouvian wedges associated with banks' investment and leverage decisions.

¹¹Risk-weights are used in practice to adjust the right-hand side of this constraint for different types of risky investments. We abstract from risk-weights because we focus on a single type of risky investment.

Table 1: Direct Measurement

Leakage/Policy Elasticities	Variable	Value
Traditional bank investment	$\frac{d \log k_0^R}{d \tau_b^R}$	-5.3335
Traditional bank leverage	$\frac{db^R}{d \tau_b^R}$	-1
Shadow bank investment	$\frac{d \log k_0^U}{d \tau_b^R}$	3.9860
Shadow bank leverage	$\frac{db^U}{d \tau_b^R}$	$\simeq 0$

Note: A detailed description of our calculations and assumptions is given in the text. Elasticities are based on estimates in [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) and [Jiang, Matvos, Piskorski, and Seru \(2020\)](#).

To quantify the welfare effects in Equation (13), we use direct measurements of leakage elasticities and market shares in the context of the US market for residential mortgages. We employ estimates from [Buchak, Matvos, Piskorski, and Seru \(2018\)](#), summarized in Table 1, who measure the responses of traditional and shadow banks’ mortgage origination to changes in traditional banks’ required equity/asset ratio. Because the empirical setting uses quantity constraints, the regulatory change maps into a tax change $d\tau_b^R$ that increases the required equity/asset ratio $\theta^R = 1 - b^R$ of traditional banks by one marginal unit or, conversely, reduces b^R by one marginal unit. We therefore normalize $\frac{db^R}{d\tau_b^R} = -1$ in Equation (13).¹² The leakage elasticities $\frac{d \log k_0^i}{d \tau_b^R}$ estimated by [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) imply that this tax change raises shadow bank loan investment by about 4% and lowers traditional bank investment by about 5%.¹³ We also need an estimate for the leakage elasticity $\frac{db^U}{d \tau_b^R}$, which is challenging because shadow bank leverage b^U is not recorded in standard administrative datasets. Using recently collected data on b^U , [Jiang, Matvos, Piskorski, and Seru \(2020\)](#) find no clear time trend in unregulated lenders’ leverage between 2011 and 2017 — a period during which traditional bank regulation became much tighter, and shadow banks dramatically grew their market share — and that shadow banks’ leverage is mostly explained by firm fixed effects such as size, rather

¹²More rigorously, we scale taxes by $\chi = -\frac{db^R}{d\tau_b^R}$ to impose (locally) that $\frac{db^R}{d\tau_b^R} = -1$. All empirical leakage elasticities reported in this section are in terms of scaled taxes.

¹³The semi-elasticities $\frac{d \log k_0^i}{d \tau_b^R}$ can be used in Equation (13) to obtain $\frac{dW}{d\tau_b^R} = \sum_{i \in \{R, U\}} \left(\omega_k^i \frac{d \log k_0^i}{d \tau_b^R} + \omega_b^i \frac{db^i}{d \tau_b^R} \right) k_0^i$. We derive an exact mapping between the regression results in [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) and these elasticities in Appendix D.

than interactions with traditional bank regulation.¹⁴ We therefore use $\frac{db^U}{d\tau_b^R} \simeq 0$ as our baseline, and report a sensitivity analysis in Appendix D.2.

Using these estimates, we report welfare effects as a function of three Pigouvian wedges ω_k^R , ω_b^R , and ω_k^U , as well as the market share of shadow banks.¹⁵ We parametrize our illustration by defining $\lambda = \omega_k^U/\omega_k^R \times k_0^U/k_0^R$, which summarizes the relative distortion associated with shadow and traditional banks, times their relative market share. In Figure 1, we plot the contours of the welfare effect in Equation (13), as a function of ω_k^R (horizontal axis) and ω_b^R (vertical axis), and for different values of λ (across the three panels). The thick dashed line in each panel delineates the points where $\frac{dW}{d\tau_b^R} = 0$. Lighter shaded areas delineate the region where $\frac{dW}{d\tau_b^R}$ is positive, implying that tighter leverage regulation would locally improve welfare, while looser regulation is welfare-improving in the darker areas.

The left panel in Figure 1 is a benchmark in which shadow banks impose no relevant distortions ($\lambda = 0$). In this case, leakage to shadow banks is not costly, but leakage from leverage regulation to traditional bank investment still matters because investment remains unregulated. Since the leakage elasticity $\frac{d \log k_0^R}{d\tau_b^R} < 0$ in the data, traditional bank leverage and investment are complements empirically. Our theoretical results suggest that it is optimal to overregulate traditional banks' leverage, as long as the planner considers investment to be underregulated ($\omega_k^R < 0$). Consistent with this intuition, the plot shows that tighter leverage regulation tends to improve welfare in this region. However, this conclusion changes when traditional bank leverage is already significantly overregulated relative to first-best ($\omega_b^R \gg 0$).

In the middle panel in Figure 1, traditional and shadow banks are symmetric ($\lambda = 1$). Welfare effects now reflect a tradeoff between i) the complementarity captured by $\frac{d \log k_0^R}{d\tau_b^R} < 0$, which pushes for overregulation, and ii) the substitutability between traditional bank leverage and shadow bank investment via $\frac{d \log k_0^U}{d\tau_b^R} > 0$, which pushes for underregulation. Empirically, the first effect tends to dominate. As before, tighter leverage regulation tends to improve welfare when investment is underregulated.

¹⁴In addition, the typical need for leverage among shadow banks in the US mortgage market arises from warehouse lines of credit, which provide short-term bridge financing for loans that shadow banks plan to sell in secondary markets — see Jiang (2023). The proportion of shadow bank originations sold is also roughly constant over time, again suggesting that their leverage decisions are independent of traditional banks' leverage regulation.

¹⁵The numerical range for Pigouvian wedges we consider is an interval around the range of plausible quantitative estimates from our equilibrium model below. In our baseline scenario, the wedge ω_b^U associated with shadow banks' leverage does not affect welfare.

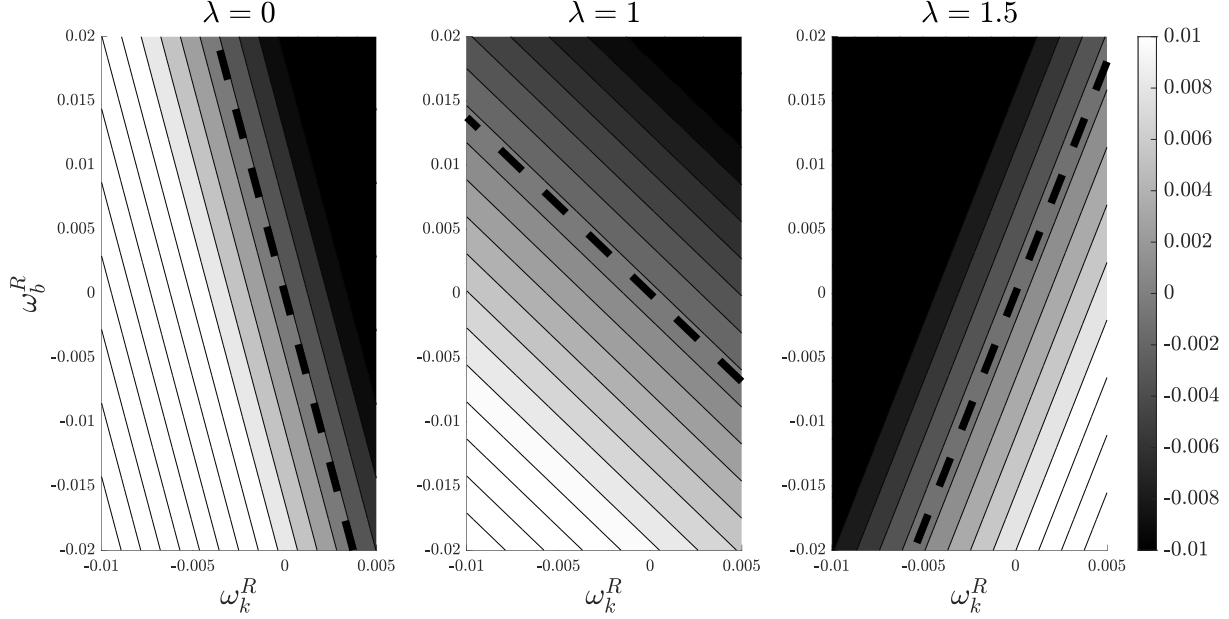


Figure 1: Marginal Welfare Effects Based on Direct Measurement

Note: This figure plots $\frac{1}{k_0^R} \frac{dW}{d\tau_b^R}$, the marginal welfare effect of tighter leverage regulation scaled by traditional bank investment, as a function of Pigouvian wedges $\{\omega_k^R, \omega_k^U, \omega_b^R\}$ using the empirical estimates for leakage elasticities from Table 1, setting $\lambda = \omega_k^U / \omega_k^R \times k_0^U / k_0^R$. The thick dashed lines show combinations of wedges for which $\frac{dW}{d\tau_b^R} = 0$.

In the right panel in Figure 1, shadow banks impose larger distortions than traditional banks and/or have greater market share ($\lambda = 1.5$). Now leakage to shadow bank investment dominates, so tighter leverage regulation tends to reduce welfare (and, by symmetry, looser regulation would improve welfare), unless leverage is already significantly underregulated.

All panels highlight the role of the investment wedge ω_k^R and how it interacts with λ . When $\lambda = 0$, tighter leverage regulation improves welfare if ω_k^R is *large* in absolute value, by crowding out investment since $\omega_k^R < 0$ means that investment is currently excessive. When $\lambda = 1$, the welfare conclusions are less sensitive to the value of ω_k^R . When shadow banks impose large marginal distortions or have greater market share — with $\lambda = 1.5$ — tighter leverage regulation improves welfare if ω_k^R is *small* in absolute value. This is because the benefits from tightening traditional banks' leverage are offset by an investment increase by shadow banks, which is associated with a large distortion.

The exercise shows how measured leakages convert Pigouvian wedges into welfare statements. One could try to measure the relevant Pigouvian wedges directly, for

example, by leveraging statistical measures of systemic risk in banking (e.g., [Adrian and Brunnermeier, 2016](#)). However, our preferred approach is to combine direct measurement of leakage elasticities with a quantitative model that generates Pigouvian wedges. We turn to this approach in the next section.

4.2 Model Quantification

We now present a simple but fully specified equilibrium model, which illustrates how regulatory constraints quantitatively impact the optimal constrained regulation. Our model stays as close as possible to the general framework outlined above, while explicitly capturing three distinct distortions/wedges that are central to the design of financial regulation: i) bailouts, ii) pecuniary externalities, and iii) distorted beliefs. We keep the description of the model in this section as concise as possible. For a complete and rigorous definition of preferences and technologies, as well as a detailed solution, we refer the reader to [Appendix E](#).

Environment. There are two types of investors, regulated banks R and unregulated banks U , as well as creditors C . Investors discount the future (i.e., dates 1 and 2) at rate β , while creditors' discount factor is $\beta^C > \beta$ — their relative patience generates gains from trade. There are three dates $t \in \{0, 1, 2\}$. We first explain the timing of events and resolution of uncertainty, which is also illustrated in [Figure 2](#).

- At $t = 0$, investors make investment and leverage decisions $\mathbf{x}^i = (k_0^i, b^i)$. They purchase k_0^i units of durable investment/capital goods at a competitive price p_0 . Investments generate a potential return v per unit, and incur a convex adjustment cost $\Psi^i(k_0^i)$. Moreover, they issue debt to creditors, with face value $b^i k_0^i$, that allows them to raise $Q^i(b^i) k_0^i$ at date 0, where $Q^i(\cdot)$ — as in the baseline model — is an endogenous pricing schedule.¹⁶ Investors pay corrective taxes on $\mathbf{x}^i$.
- At $t = 1$, a fraction ζ^i of intermediaries' debt must be repaid. Moreover, a binary shock $z \in \{0, 1\}$ is realized. In the event $z = 0$, which stands for normal times, investors receive early returns $\eta^i v$ per unit of investment, which can be used to repay debt. In the event $z = 1$, which stands for a financial crisis, investments do not yield any early returns. In this case, investors must fire-sell a fraction of their

¹⁶In the terminology of [Dubey, Geanakoplos, and Shubik \(2005\)](#), the schedule $Q^i(\cdot)$ is a credit surface.

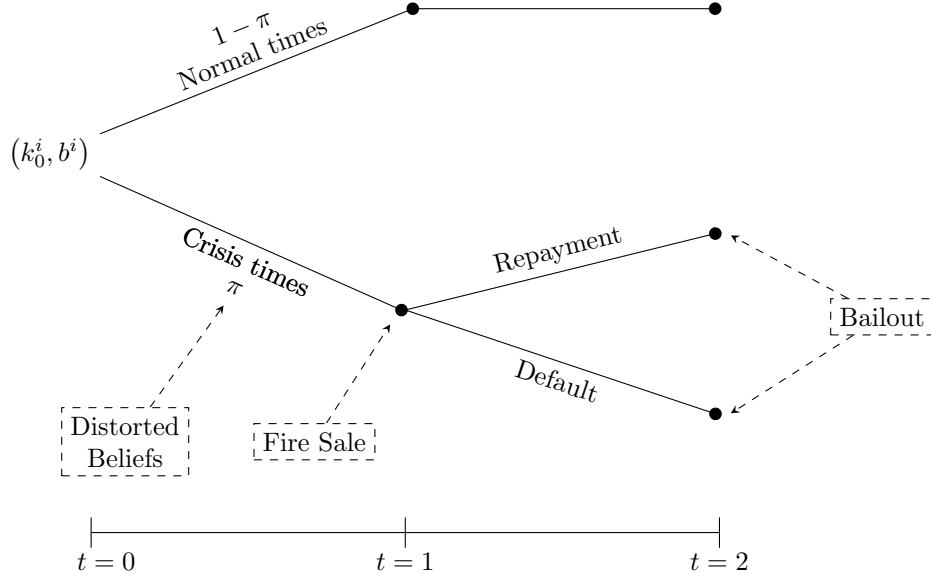


Figure 2: Model Timing

Note: This figure illustrates the timing of the equilibrium model used in Section 4.2. Crisis times occur with probability π at date 1, after banks have made investment and leverage decisions, k_0^i and b^i respectively, at date 0. Distorted beliefs, fire sales, and bailouts are the three rationales for corrective regulation in this model.

investments at a competitive price p_1 to meet repayment needs.¹⁷ We write θ^i for the fraction of investments that is retained (i.e., not fire-sold) in a crisis event, which we characterize in the Appendix.

- At $t = 2$, the remaining fraction $1 - \zeta^i$ of debt must be repaid. In normal times, investments yield their remaining return $(1 - \eta^i)v$. In a crisis, investments that have not been fire-sold yield a cash flow of $(1 - \eta^i)v + s$, where s is a continuous random variable. Investors also get a government bailout $t^i(b^i, s)$ per unit of initial investment. The government funds bailouts by imposing a tax $(1 + \kappa)t^i(b^i, s)$ on creditors, where κ captures deadweight fiscal costs. There are now two possible scenarios:

- Default: If cash flows including bailouts are insufficient to cover remaining debt, then the type i investor is in default.¹⁸ In this case, creditors seize bailouts

¹⁷Concretely, a fire sale becomes necessary if $\eta^i v < \zeta^i b^i$. We assume throughout that η^i is large enough to rule out a fire sale in normal times.

¹⁸The formal condition for default is $(1 - \zeta^i)b^i > \theta^i[(1 - \eta^i)v + s] + t^i(b^i, s)$. This typically implies the existence of a boundary state $\hat{s}$, such that default occurs if and only if $s < \hat{s}$. We characterize $\hat{s}$ explicitly in the appendix.

and a fraction ϕ of investment returns, so that $1 - \phi$ stands for the deadweight cost of default.

- Solvency: If cash flows are large enough to repay the remaining debt, then investors are solvent, repay their debt, and consume any residual claim.

We next specify probabilistic beliefs. All agents in the model perceive the probability of crisis at date 1 to be $\Pr[z = 1] = \pi$, while the planner perceives it to be π^P . When $\pi \neq \pi^P$, the planner therefore views agents' beliefs as distorted, and designs corrective policy partially to counteract this “internality”. Moreover, the planner and agents agree on the distribution s of residual uncertainty at date 2. We write $\mathbb{E}[\cdot]$ for (common) expectations over s .

Finally, we close the model with a population of outside investors, whose role is to clear the primary and secondary markets for investments. At date 0, outsiders can produce investment goods at a convex cost $\Upsilon(k)$ and sell them to investors. This technology pins down the price of capital $p_0 = \Upsilon'(\sum_i k_0^i)$ in equilibrium. At date 1, in case of a crisis ($z = 1$), outsiders can buy fire-sold investments from investors and extract a concave liquidation payoff $H(k)$. This technology pins down the equilibrium fire-sale price $p_1 = H'(\sum_i (1 - \theta^i) k_0^i)$.

Summary of Equilibrium Behavior. At date 0, creditors offer banks a credit surface of the following form:

$$Q^i(b^i) = \beta^C \left[\zeta^i b^i + (1 - \zeta^i) \left((1 - \pi) b^i + \pi \mathbb{E}[\mathcal{R}^i(s)] \right) \right].$$

This is the risk-neutral valuation of debt repayments. A fraction ζ^i of debt is always repaid at date 1. At date 2, the remaining debt is repaid in full in normal times, which occur with probability $1 - \pi$. In crises, the repayment may be reduced in the event of default, as captured by a state-contingent repayment function $\mathcal{R}^i(s)$. In the Appendix, we rigorously derive $\mathcal{R}^i(s)$, and further show that the optimality conditions for optimal leverage b^i and investment k_0^i are given by

$$M^i(b^i, p_1) = p_0 + \Psi^{i'}(k_0^i) + \tau_k^i \tag{14}$$

$$\frac{\partial M^i(b^i, p_1)}{\partial b^i} = \tau_b^i, \tag{15}$$

where the function $M^i(\cdot)$ corresponds to the joint valuation of the (inside) equity and debt claims issued by type i banks. Notice that this valuation depends on the endogenous price p_1 , at which (shadow) banks can fire-sell assets in the event of a crisis. Therefore, both the initial capital price p_0 and the fire-sale price p_1 generate endogenous linkages between banks and shadow banks, which will drive the key leakage elasticities that we consider below.¹⁹

While banks and shadow banks use the valuation function $M^i(\cdot)$ to make decisions, which includes their assessment π of the probability of crisis, recall that the planner considers a different probability of crisis π^P to be true. Therefore, the planner evaluates investment and leverage decisions according to an analogous function $M^{i,P}(\cdot)$, which replaces π with π^P , and is defined rigorously in the Appendix.

Policy/Leakage Elasticities. As in our general theory, policy elasticities are key objects for optimal policy. We solve for these elasticities numerically below, but it is possible to provide useful analytical guidance. To understand these effects, it helps to first ignore the endogeneity of fire sales, and treat the fire-sale price p_1 as a constant. We then have the following result:

Lemma 2. *(Quantitative Model: Leakage Elasticities) The system of first-order conditions in (14) and (15) has the following comparative statics for a fixed fire-sale price p_1 :*

1. *If banks face a higher leverage regulation τ_b^R , then they decrease their leverage and investment, while shadow banks leave their leverage unchanged and increase their investment.*
2. *If banks face a higher capital regulation τ_k^R , then they decrease their investment, while shadow banks increase their investment. Both types' leverage is unchanged.*

Lemma 2 shows that, holding fire-sale prices constant, leverage and capital are complements within banks: an increase in τ_b^R decreases both bank leverage and investment. Moreover, across institutions, bank decisions and shadow bank investment are substitutes: an increase in either τ_b^R or τ_k^R increases shadow bank investment. This channel operates through the market price p_0 of investment capital at date 0. Decreased

¹⁹It is possible to introduce further linkages via creditors' demand for debt, for example, by making creditors risk-averse — as in an earlier version of this paper — by introducing nonpecuniary preferences for debt holdings, or through direct linkages, for example, with banks providing funding to shadow banks.

demand due to tighter regulation on banks lowers the equilibrium price of investments, thus incentivizing shadow banks to invest more. Lemma 2 also makes clear that leverage only responds to direct taxation. For instance, bank regulations do not affect shadow bank leverage for a given level p_1 of fire-sale prices.²⁰

Once we take into account the equilibrium response of fire-sale prices, the general comparative statics with respect to regulations are ambiguous. However, a strong intuition is that endogenous fire sales introduce additional substitutability. Any decrease in bank investment or leverage, for instance, reduces the severity of the fire sale at date 1, thereby giving shadow banks a greater incentive to invest and take leverage. In our calibration below, these effects are quantitatively small, and the dominant patterns are those described in Lemma 2.

Marginal Distortions and Welfare Effects. In this model, as explained above, marginal welfare effects of policy changes are given by Equation (13), which applies our general Lemma 1 in the context of this application.²¹ To operationalize this equation, the next result states the marginal distortions (and, hence, Pigouvian wedges) that are implied by the quantitative model.

Lemma 3. (*Quantitative Model: Pigouvian Wedges*) *The marginal distortions $\delta = \{\delta_b^i, \delta_k^i\}_{i \in \{R, U\}}$ associated with investors' decisions in the quantitative model are defined as follows:*

$$\delta_b^i = \underbrace{(1 + \kappa) \beta^C \pi^P \mathbb{E} \left[\frac{\partial t^i(b^i, s)}{\partial b^i} \right]}_{\text{Bailouts}} + \underbrace{\pi^P \delta_p \frac{\partial p_1}{\partial k_0^i}}_{\text{Pecuniary}} + \underbrace{\frac{\partial M^i(b^i, p_1)}{\partial b^i} - \frac{\partial M^{i,P}(b^i, p_1)}{\partial b^i}}_{\text{Distorted beliefs}} \quad (16)$$

$$\delta_k^i = \underbrace{(1 + \kappa) \beta^C \pi^P \mathbb{E} \left[t^i(b^i, s) \right]}_{\text{Bailouts}} + \underbrace{\pi^P \delta_p \frac{\partial p_1}{\partial k_0^i}}_{\text{Pecuniary}} + \underbrace{M^i(b^i, p_1) - M^{i,P}(b^i, p_1)}_{\text{Distorted beliefs}}, \quad (17)$$

where $\frac{\partial p_1}{\partial b^i}$ and $\frac{\partial p_1}{\partial k_0^i}$ denote the general equilibrium sensitivity of fire-sale prices to changes in leverage and investment; and where δ_p , defined in the Appendix, corresponds to a marginal pecuniary distortion. Given a set of taxes/regulations $\tau = \{\tau_b^i, \tau_k^i\}_{i \in \{R, U\}}$, the Pigouvian wedges are $\omega = \tau - \delta$, and the marginal welfare effects of changing regulations are given

²⁰Below, we also consider several scenarios in which shadow banks can be regulated. In that case, the effects of higher regulations on shadow banks, τ_b^U and τ_k^U , are symmetric to those described in Lemma 2.

²¹We also derive Equation (13) more explicitly, in terms of the primitives of the quantitative model, in the Appendix.

by Equation (13).

The first terms in Equations (16) and (17) capture the expected value of the *bailout*-induced externality that banks' decisions impose on creditors in a crisis, who ultimately fund the bailout, augmented by the fiscal deadweight cost κ . The second terms account for *pecuniary* externalities that banks' decisions impose on other banks and outsiders via prices in fire-sale events. The key object in this term is δ_p , which heuristically measures the sensitivity of social surplus to prices in the event of a crisis, and which we define in the Appendix. Intuitively, higher prices induce a distributive pecuniary externality (Dávila and Korinek, 2018) by shifting wealth to constrained investors. This is due to the incompleteness of insurance markets. In addition, higher prices move the default boundary, giving rise to a novel frictional pecuniary externality. Our formal definition of δ_p in the Appendix includes both types of pecuniary effect. Finally, the third terms in Equations (16) and (17) accounts for the planner's desire to correct perceived *belief distortions*, or "internalities". In our setting, the planner finds it desirable to regulate decisions whenever agents' and planner's beliefs differ.

4.2.1 Calibration

For quantification, we assume that the residual cash flow risk in a crisis follows $s \sim \mathcal{N}(0, \sigma^2)$. We make investors' adjustment cost functions isoelastic, $\Psi^i(k_0^i) = \psi^i \frac{(k_0^i)^{\nu^i}}{\nu^i}$, and outsiders' cost function quadratic, with $\Upsilon(k) = \frac{1}{2}k^2$. Outsiders' date 1 technology takes a shifted isoelastic form $H(k) = \frac{(1+k)^\gamma}{\gamma}$. Finally, we assume that the bailouts in a crisis are linear functions of leverage, $t^i(b^i, s) = \alpha_0^i + \alpha_b^i b^i$. This functional form approximates more general models of government bailouts without commitment.

Table 2 summarizes our baseline parametrization. Interpreting the time between date 0 and dates 1 (and 2) as a year, we set a discount factor for creditors of 0.98, consistent with a 2% risk-free interest rate, and set $\beta = 0.972$ for other agents, targeting an average leverage (debt to assets) of 0.825. We set agents' perceived crisis probability to $\pi = 0.03$ (a crisis roughly every 33 years), and the planner's probability to $\pi^P = 0.05$, consistent with Reinhart and Rogoff (2009). We set $\zeta^R = 0.2$ and $\zeta^U = 0.3$ to target differences in leverage ratios between regulated and shadow banks: 0.86 and 0.79, respectively, within the range reported in Jiang, Matvos, Piskorski, and Seru (2020). This choice implies that a quarter of liabilities are on average due at date 1. By imposing that $\eta^i = \zeta^i$ we ensure that the timing of investment payoffs and liabilities is aligned in normal times, ensuring

Table 2: Parameter Values

Description		Parameter(s)	Value(s)
Preferences	Discount Factor Banks/Outsiders	β	0.972
	Discount Factor Creditors	β^C	0.98
Technology (Profitability)	Probability of Crisis Times	π, π^P	0.03, 0.05
	Share of Early Payoff	η^R, η^U	0.2, 0.3
	Investment Payoff	v	1.11
	Payoff Dispersion	σ	0.1
Technology (Costs)	Traditional Banks' Adjustment Cost	ψ^R, ν^R	0.0058, 1.19
	Shadow Banks' Adjustment Cost	ψ^U, ν^U	0.0005, 2
	Outsiders' Technology	γ	0.15
Financing	Recovery after Default	ϕ	0.72
	Share of Early Debt	ζ^R, ζ^U	0.2, 0.3
Bailout	Bailout Generosity	$\alpha_0^R, \alpha_0^U, \alpha_b^R, \alpha_b^U$	0.033, 0.022, 0.1095, 0.073
	Marginal Cost of Public Funds	κ	0.15

Note: This table summarizes the choice of parameter values used in Section 4.2.

that no fire sales occur. By setting $v = 1.11$, the model matches a return on average equity of around 8%, while a value of the return dispersion $\sigma = 0.1$ ensures that failure probabilities for traditional banks are 3.8%, slightly larger than historical averages but consistent with implied default probabilities extracted from CDS spreads.

The banks' adjustment cost parameters ψ^R , ψ^U , ν^R , and ν^U are jointly chosen to match the leakage elasticities reported in Table 1 — and used in our direct measurement exercise — and to ensure that the market share of shadow banks is 57%, consistent with current estimates reported in Jiang (2023) and Buchak, Matvos, Piskorski, and Seru (2024a). By setting the curvature of outsiders' technology to $\gamma = 0.15$, we can target a sizable drop in prices in a fire sale of roughly 20%. We set the recovery parameter $\phi = 0.72$ to match a 28% average default deadweight loss, consistent with estimates from Granja, Matvos, and Seru (2017). To capture differences in government guarantees between traditional and shadow banks, we impose that bailout parameters are larger for traditional banks by a factor of 1.5. We set $\alpha_0^R = 0.033$ and $\alpha_b^R = 0.1095$ so that the expected implied bailout to equity ratio is roughly 1.5% for traditional banks, where roughly 25% of the bailout is untargted (so 75% is conditional on b). Based on Dahlby (2008), we set the net marginal cost of public funds to $\kappa = 0.15$.

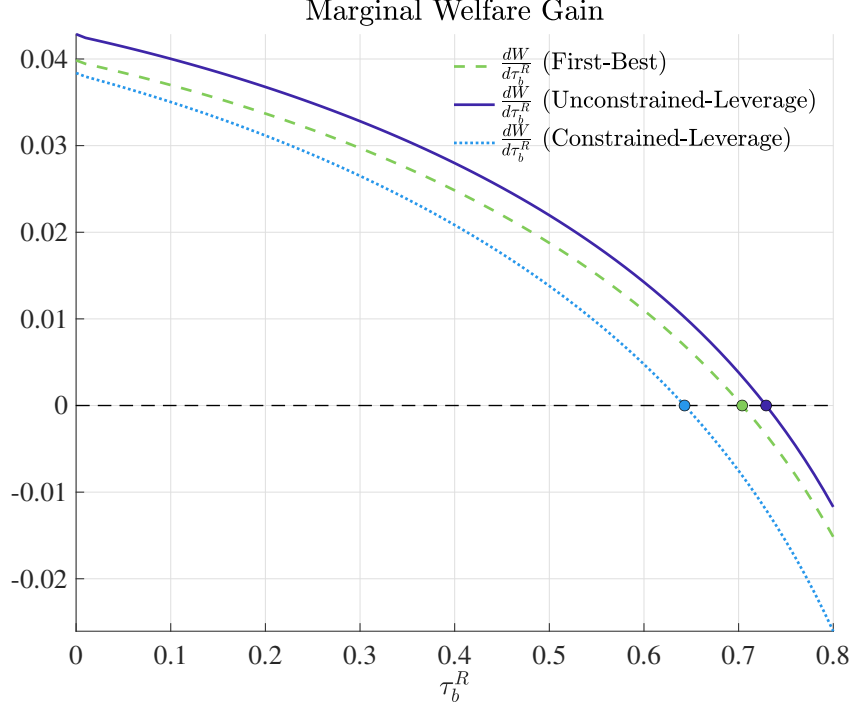


Figure 3: First-Best vs. Unconstrained-Leverage-Regulation vs. Constrained-Leverage-Regulation

Note: This figure shows the marginal welfare gain of adjusting the leverage regulation on R banks for three different scenarios. The green dashed line corresponds to the first-best scenario, in which all decisions are perfectly regulated. The solid blue line corresponds to the unconstrained-leverage-regulation scenario, in which investment regulations are constrained to be zero. The dotted light blue line corresponds to the constrained-leverage-regulation scenario, in which the only available regulation is on the leverage decision of R banks. In the first-best and unconstrained-leverage-regulation scenarios, the remaining unconstrained regulations are held constant at their optimal value. The regulation τ_b^R is reported in percentage points.

We calibrate the model using a scenario in which the planner can regulate only the leverage decision of traditional banks, while the leverage decision of shadow banks, as well as investment decisions, remain unregulated. This scenario, within our model framework, is closest to current financial regulation.

4.2.2 Optimal Policy: Quantitative Analysis

We can now use the calibrated model to explore the quantitative impact of different regulatory constraints on the optimal regulation. Table 3 summarizes our quantitative results, which we describe next. Figures 4 and 5 illustrate the marginal welfare effects associated with each scenario.

We start with the first-best benchmark, summarized in the first row of Table 3.

Table 3: Optimal Policies

Regulatory Scenario	Optimal Regulation				Leverage		Investment	
	τ_b^R	τ_b^U	τ_k^R	τ_k^U	b^R	b^U	k_0^R	k_0^U
First Best	0.704	0.588	0.959	0.949	0.855	0.749	0.455	0.613
Unconstrained Leverage	0.729	0.577	0	0	0.851	0.750	0.482	0.595
Constrained Leverage	0.643	0	0	0	0.860	0.790	0.461	0.616
Uniform Leverage	0.650		0	0	0.862	0.743	0.529	0.549

Note: All regulations, τ_b^R , τ_b^U , τ_k^R , and τ_k^U , are reported in percentage points.

The optimal regulation perfectly targets the (strictly positive) marginal distortions characterized in Lemma 3. Traditional banks' leverage decisions are more tightly regulated due to larger bailout-induced marginal distortions in our baseline calibration.

The optimal taxes in the first-best case also show marginal distortions. Indeed, the first-best planner sets $\tau^* = \delta$ in line with the Pigou principle. The first row of Table 3 therefore shows that banks impose larger leverage distortions δ_b^i than shadow banks, and slightly larger investment distortions δ_k^i . This occurs because banks take substantially more leverage — emboldened by the anticipation of larger bailouts. Nonetheless, the distortions of shadow banks are still sizable because, in our calibration, their debt capacity is more unstable in a crisis, making them larger marginal contributors to fire sales.

Second, we model an *unconstrained-leverage-regulation* scenario in which both banks and shadow banks are regulated, but only via the leverage regulation τ_b^i . This reflects practical instruments like capital requirements, which — as discussed above — leave the scale k_0^i of investment as an unregulated decision. The optimal regulation *overregulates* the leverage decisions of traditional banks but *underregulates* the leverage decisions of shadow banks relative to the first-best. Proposition 1 provides clear intuition: The planner views the (risky) investment of banks and shadow banks as underregulated. When regulating banks' leverage, the regulator trades off increasing τ_b^R to curb banks' investment (a complement to their leverage) against decreasing τ_b^R to curb shadow banks' investment (a substitute). The complementarity dominates for banks, whose investment is associated with slightly larger Pigouvian wedges, and hence the optimal policy is to overregulate bank leverage. Conversely, for shadow banks, the substitutability dominates, and the planner underregulates.

Third, we model a *constrained-leverage-regulation* scenario, in which the only regulated decision is bank leverage b^R . Relative to the previous scenario, the planner now becomes

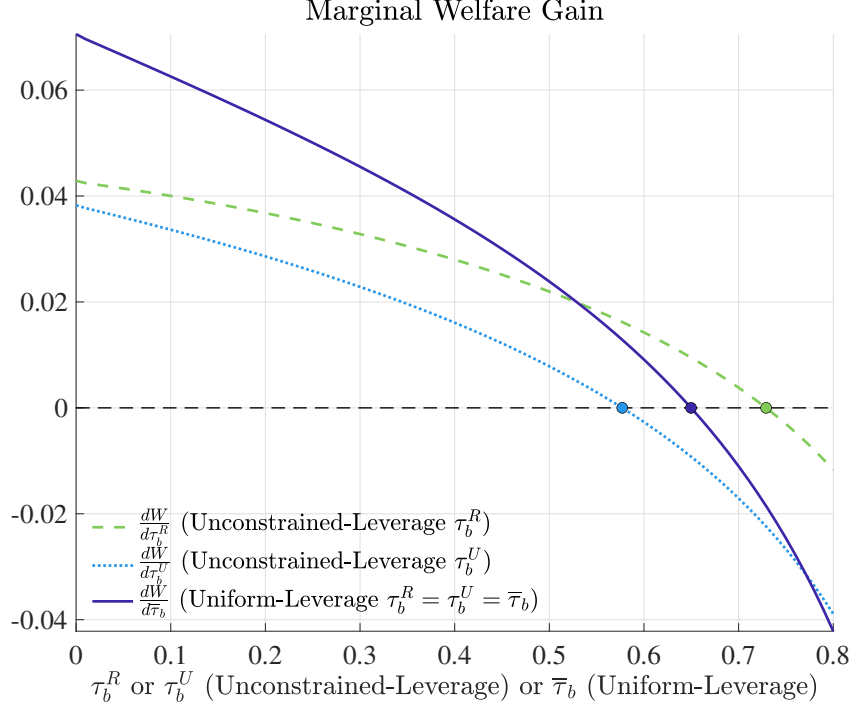


Figure 4: Uniform-Leverage-Regulation

Note: This figure shows the marginal welfare gain of adjusting different regulatory instruments to compare the unconstrained-leverage-regulation scenario with the uniform-leverage-regulation scenario. The green dashed line and the dotted blue line correspond to the marginal welfare gain associated with independently varying τ_b^R and τ_b^U , respectively, while the other is held at the optimal value in the unconstrained-leverage-regulation scenario. The solid blue line corresponds to the marginal welfare gain associated with adjusting the uniform regulation $\tau_b^R = \tau_b^U = \bar{\tau}_b$. Consistent with Equation (11), the optimal uniform regulation is a weighted average of the regulation the planner would set if chosen freely. The regulations are reported in percentage points.

more concerned about leakage to shadow bank investment, a substitute to b^R , and therefore switches to underregulating bank leverage relative to first-best. In addition, the planner becomes concerned about leakage to shadow bank leverage b^U , which is another substitute to b^R (via the response of fire-sale prices p_1) and pushes further towards underregulation. This scenario is arguably closest to current regulatory practice, in which banks are heavily regulated via leverage/capital requirements, and shadow banks are mostly unregulated.

Finally, we consider a uniform-leverage-regulation scenario, in which both banks' and shadow banks' leverage must be subject to the same regime: $\tau_b^R = \tau_b^U = \bar{\tau}_b$. This allows us to evaluate simple policy reforms that would bring shadow banks into the same regulatory perimeter as traditional banks. As anticipated by our general results in Section 3, the planner sets the uniform regulation as a weighted average of bank and shadow bank

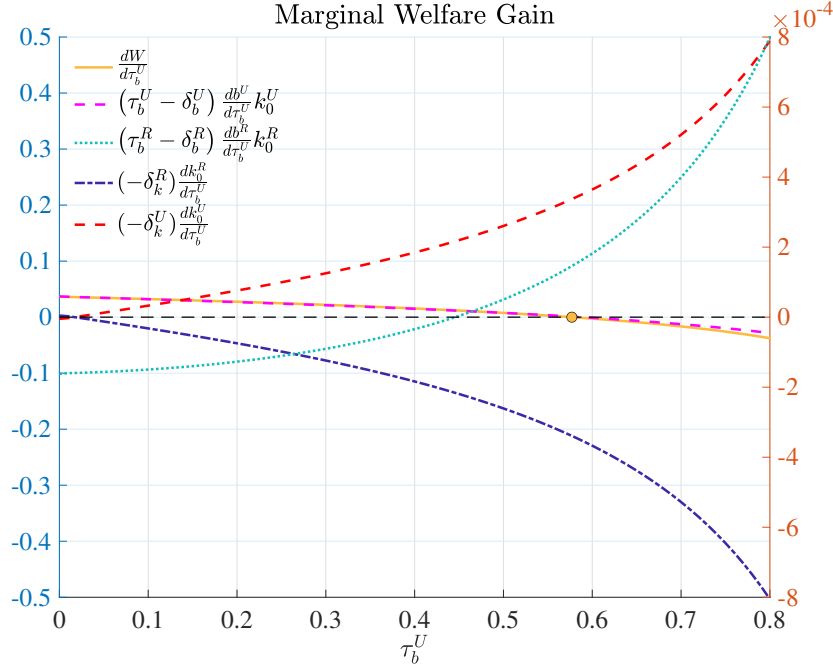


Figure 5: Relaxing Constraint on Leverage Regulation

Note: This figure shows the marginal welfare gains of increasing the leverage regulation on shadow banks starting from the constrained-leverage-regulation scenario, in which $\tau_b^U = 0$. The welfare gains of increasing τ_b^U are positive and decreasing. The direct effect of increasing τ_b^U is larger than the overall effect, consistent with our theoretical results. The reverse leakage effect is negative, attenuating the direct gains from regulation. The dotted green line $(\tau_b^R - \delta_b^R) \frac{db^R}{d\tau_b^U} k_0^R$ is plotted on the right axis. The regulation τ_b^U is reported in percentage points.

distortions — weighted by the relevant elasticities and adjusted for reverse leakage from unregulated scale decisions, as in Equation (11).

Our results in this application highlight that leakage elasticities featuring both substitutability and complementarity naturally emerge in common regulatory scenarios, so a quantitative analysis that incorporates different banks' different decisions is necessary. Quantitatively, Table 3 shows that the leverage ratio of shadow banks in the constrained-leverage-regulation scenario is 0.79, while it would be roughly 0.75 in the first-best and unconstrained-leverage-regulation scenarios, a substantial difference. The difference in shadow bank investment is somewhat smaller but also significant, going from 0.616 to 0.595, or roughly a 3.5% reduction. This reduction is partly compensated by a roughly 4.5% increase in traditional banks' investment, from 0.461 to 0.482.

In order to shed further light on policies that bring shadow banks under regulatory control, we consider marginal policy changes. The planner starts from the constrained-

leverage-regulation scenario (i.e., the one closest to current practice). In this scenario, shadow bank leverage cannot be regulated; the key regulatory constraint is $\tau_b^U \leq 0$. Then, we gradually relax the constraint on τ_b^U until it becomes slack. This exercise illustrates the effects of relaxing regulatory constraints and demonstrates the role played by leakage and reverse leakage elasticities. Figure 5 provides an illustration of our results.

The solid yellow line shows the overall marginal welfare effect, which is initially positive, but approaches zero as τ_b^U approaches its optimum value. However, this well-behaved overall welfare gain masks multiple effects, as we explain next. Formally, we show in the appendix that the marginal welfare gain of increasing τ_b^U can be expressed as

$$\frac{dW}{d\tau_b^U} = (\tau_b^U - \delta_b^U) \underbrace{\frac{db^U}{d\tau_b^U} k_0^U}_{<0} + (\tau_b^R - \delta_b^R) \underbrace{\frac{db^R}{d\tau_b^U} k_0^R}_{\gtrsim 0} - \delta_k^U \underbrace{\frac{dk_0^U}{d\tau_b^U}}_{<0} - \delta_k^R \underbrace{\frac{dk_0^R}{d\tau_b^U}}_{>0}.$$

This marginal gain consists of four terms. First, there is the immediate direct effect of tightening leverage regulation on shadow banks, reducing their leverage ($\frac{db^U}{d\tau_b^U} < 0$). Since shadow banks are underregulated, this direct effect contributes positively to $\frac{dW}{d\tau_b^U}$, and is quantitatively important. Second, there is the reverse leakage effect on the leverage of traditional banks. In our calibration, it turns out that $\frac{db^R}{d\tau_b^U}$ is slightly positive but almost negligible, which explains why we plot it using a different vertical axis. Since the Pigouvian wedge $\tau_b^R - \delta_b^R$ switches from negative (underregulation) to positive (overregulation) as the economy transitions from the constrained-leverage-regulation scenario to the unconstrained-leverage-regulation scenario, this term initially contributes negatively to $\frac{dW}{d\tau_b^U}$ but eventually becomes positive.²² Third, there is the impact on shadow banks' investment, which is a gross complement as explained above, so $\frac{dk_0^U}{d\tau_b^U} < 0$. Since shadow banks' investment is underregulated, this effect also contributes positively to $\frac{dW}{d\tau_b^U}$. Finally, there is the impact on traditional banks' investment, which in this model is the key source of substitutability among banks' decisions, so $\frac{dk_0^R}{d\tau_b^U} > 0$. Since traditional banks' investment is underregulated, this effect contributes negatively to $\frac{dW}{d\tau_b^U}$. Quantitatively, the last two effects working in opposite directions roughly cancel, so that the overall effect is almost exclusively driven by the immediate direct effect of tightening leverage regulation on shadow banks.

This analysis illustrates the direct effects of relaxing constraints on regulation, as well

²²In the Online Appendix, we show how to express $\frac{dW}{d\tau_b^U}$ in terms of elements of the Le Chatelier matrix L introduced in Section 3.

as the attenuation of these effects due to reverse leakage, both of which are part of our general characterization in Proposition 2. As in our general theory, reverse leakage effects attenuate the direct effects, in particular by pushing distortive activities back into the regulated system, which is underregulated. Overall, the role of attenuation is relatively weak in this application, and thus our model implies that there are welfare gains from regulating shadow banks.

5 Conclusion

This paper has studied optimal financial corrective regulation when policy instruments are imperfect, a question of first-order importance for regulators and policymakers. We have shown that leakage elasticities from perfectly regulated to imperfectly regulated decisions, along with Pigouvian wedges, determine the optimal regulation of perfectly regulated decisions. Notably, the optimal second-best policy hinges on whether perfectly and imperfectly regulated decisions are gross substitutes or complements. Reverse leakage elasticities from imperfectly to perfectly regulated decisions influence the optimal regulation of imperfectly regulated decisions and determine the social value of relaxing constraints on regulation.

Our quantitative application demonstrates the value of our approach to designing optimal financial regulation, an environment with notoriously imperfect regulatory instruments. In a direct measurement exercise, we present empirical counterparts of the leakage elasticities that directly inform whether it is desirable to adjust leverage regulation in the presence of unregulated shadow banks.²³ Within a quantitative model of financial regulation that encompasses multiple rationales for policy intervention, we explore how different regulatory constraints quantitatively impact the optimal regulation in the presence of unregulated shadow banks.

²³A growing empirical literature in financial economics and beyond studies leakages and unintended consequences of regulation. A broader implication of our direct measurement approach is that reduced-form estimates from this literature can be directly used to make welfare-relevant decisions for optimal constrained corrective policy independently of model details. This insight not only highlights the usefulness of existing empirical estimates for rigorous normative analysis, but also fosters a closer link between the empirical and theoretical literatures on financial intermediation and crises, which are often perceived as largely disconnected.

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ONLINE APPENDIX

Section A of this Online Appendix defines the vectors and matrices used throughout the paper, while Section B provides the proofs and derivations for Section 3. Sections D and E present additional material related to the quantitative application in Section 4.

A Matrix Definitions

Investor i 's decisions, regulations, and marginal distortions can be expressed as $N \times 1$ vectors $\mathbf{x}^i$, $\boldsymbol{\tau}^i$, and $\boldsymbol{\delta}^i$, as follows:

$$\mathbf{x}^i = \begin{pmatrix} x_1^i \\ \vdots \\ x_n^i \\ \vdots \\ x_N^i \end{pmatrix}_{N \times 1}, \quad \boldsymbol{\tau}^i = \begin{pmatrix} \tau_1^i \\ \vdots \\ \tau_n^i \\ \vdots \\ \tau_N^i \end{pmatrix}_{N \times 1}, \quad \text{and} \quad \boldsymbol{\delta}^i = \begin{pmatrix} \delta_1^i \\ \vdots \\ \delta_n^i \\ \vdots \\ \delta_N^i \end{pmatrix}_{N \times 1}.$$

These can be collected for all I types of agents in $IN \times 1$ stacked vectors, as follows:

$$\mathbf{x} = \begin{pmatrix} \mathbf{x}^1 \\ \vdots \\ \mathbf{x}^i \\ \vdots \\ \mathbf{x}^I \end{pmatrix}_{IN \times 1}, \quad \boldsymbol{\tau} = \begin{pmatrix} \boldsymbol{\tau}^1 \\ \vdots \\ \boldsymbol{\tau}^i \\ \vdots \\ \boldsymbol{\tau}^I \end{pmatrix}_{IN \times 1}, \quad \text{and} \quad \boldsymbol{\delta} = \begin{pmatrix} \boldsymbol{\delta}^1 \\ \vdots \\ \boldsymbol{\delta}^i \\ \vdots \\ \boldsymbol{\delta}^I \end{pmatrix}_{IN \times 1},$$

where the vectors $\bar{\mathbf{x}}^i$ and $\bar{\mathbf{x}}$ are defined analogously to $\mathbf{x}^i$ and $\mathbf{x}$. The marginal welfare effects of varying all regulations, $\frac{dW}{d\boldsymbol{\tau}}$, are given by

$$\frac{dW}{d\boldsymbol{\tau}} = \begin{pmatrix} \frac{dW}{d\tau^1} \\ \vdots \\ \frac{dW}{d\tau^j} \\ \vdots \\ \frac{dW}{d\tau^I} \end{pmatrix}_{IN \times 1}, \quad \text{where} \quad \frac{dW}{d\boldsymbol{\tau}^j} = \begin{pmatrix} \frac{dW}{d\tau_1^j} \\ \vdots \\ \frac{dW}{d\tau_n^j} \\ \vdots \\ \frac{dW}{d\tau_N^j} \end{pmatrix}_{N \times 1},$$

and where $\frac{dW}{d\tau_n^j}$ denotes the marginal welfare effect of varying the regulation associated with decision n by investor j . Note that

$$\frac{dW}{d\tau^j} = \frac{d\mathbf{x}}{d\tau^j} \boldsymbol{\omega} = \sum_i \frac{d\mathbf{x}^i}{d\tau^j} \omega^i = \sum_i \sum_n \frac{dx_n^i}{d\tau^j} (\tau_n^i - \delta_n^i).$$

The Jacobian matrix of decisions $\mathbf{x}$ with respect to $\boldsymbol{\tau}$, of dimension $IN \times IN$, is given by

$$\frac{d\mathbf{x}}{d\boldsymbol{\tau}} = \begin{pmatrix} \frac{dx^1}{d\tau^1} & \cdots & \frac{dx^I}{d\tau^1} \\ \vdots & \frac{dx^i}{d\tau^j} & \vdots \\ \frac{dx^1}{d\tau^I} & \cdots & \frac{dx^I}{d\tau^I} \end{pmatrix}_{IN \times IN}, \quad \text{where} \quad \frac{d\mathbf{x}^i}{d\tau^j} = \begin{pmatrix} \frac{dx_1^i}{d\tau_1^j} & \cdots & \frac{dx_N^i}{d\tau_1^j} \\ \vdots & \frac{dx_{n'}^i}{d\tau_{n'}^j} & \vdots \\ \frac{dx_1^i}{d\tau_N^j} & \cdots & \frac{dx_N^i}{d\tau_N^j} \end{pmatrix}_{N \times N}.$$

Let R and U , respectively, be the number of perfectly and imperfectly regulated decisions in the economy. The Jacobian matrix $\frac{d\mathbf{x}^U}{d\tau^R}$, of dimensions $R \times U$, can be written as

$$\frac{d\mathbf{x}^U}{d\tau^R} = \begin{pmatrix} \cdots \\ \vdots & \frac{dx_n^i}{d\tau_{n'}^j} & \vdots \\ \cdots \end{pmatrix}_{R \times U},$$

where the decisions are such that (i, n) are imperfectly regulated and (j, n') are perfectly regulated. One can similarly define $\frac{d\mathbf{x}^U}{d\tau^U}$, $\frac{d\mathbf{x}^R}{d\tau^U}$, and $\frac{d\mathbf{x}^R}{d\tau^R}$, with dimensions $U \times U$, $U \times R$, and $R \times R$ respectively, by switching the sets of coefficients. This allows us to express $\frac{d\mathbf{x}}{d\boldsymbol{\tau}}$ when needed as

$$\frac{d\mathbf{x}}{d\boldsymbol{\tau}} = \begin{pmatrix} \frac{d\mathbf{x}^R}{d\tau^R} & \frac{d\mathbf{x}^U}{d\tau^R} \\ \frac{d\mathbf{x}^R}{d\tau^U} & \frac{d\mathbf{x}^U}{d\tau^U} \end{pmatrix}_{IN \times IN}.$$

We express the Jacobian of the constraints, $\frac{d\Phi}{d\boldsymbol{\tau}}$, a matrix of dimension $IN \times M$, as follows

$$\frac{d\Phi}{d\boldsymbol{\tau}} = \begin{pmatrix} \frac{d\Phi^1}{d\tau^1} & \cdots & \frac{d\Phi^M}{d\tau^1} \\ \vdots & \frac{d\Phi^m}{d\tau^j} & \vdots \\ \frac{d\Phi^1}{d\tau^I} & \cdots & \frac{d\Phi^M}{d\tau^I} \end{pmatrix}_{IN \times M}, \quad \text{where} \quad \frac{d\Phi^m}{d\tau^j} = \begin{pmatrix} \frac{d\Phi^m}{d\tau_1^j} \\ \vdots \\ \frac{d\Phi^m}{d\tau_N^j} \end{pmatrix}_{N \times 1},$$

and the vector of Lagrange multipliers associated with the constraints, $\boldsymbol{\mu}$, by

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_M \end{pmatrix}_{M \times 1}.$$

B Proofs and Derivations: Section 3

Investors' problem: The problem solved by investor i in Lagrangian form is

$$\max_{c_0^i, \{c_1^i(s)\}, \mathbf{x}^i} \mathcal{L}^i,$$

where $\mathcal{L}^i$ is given by

$$\begin{aligned} \mathcal{L}^i = & u^i \left(c_0^i, \{c_1^i(s)\}_{s \in S}, \{\bar{\mathbf{x}}^j\}_{j \in \mathcal{I}} \right) \\ & - \lambda_0^i \left(c_0^i - n_0^i - Q^i(\mathbf{x}^i) + \Upsilon^i(\mathbf{x}^i) + \boldsymbol{\tau}^i \cdot \mathbf{x}^i - T^i \right) \\ & - \int \lambda_1^i(s) \left(c_1^i(s) - n_1^i(s) - \rho_i(\mathbf{x}^i, s) \right) dF(s), \end{aligned}$$

where λ_0^i and $\lambda_1^i(s)$ denote the Lagrange multipliers that correspond to investor i 's budget constraints. The consumption optimality conditions imply that $\lambda_0^i = \frac{\partial u^i}{\partial c_0^i}$ and $\lambda_1^i(s) dF(s) = \frac{\partial u^i}{\partial c_1^i(s)}$. The optimality conditions for investor i are given by

$$-\lambda_0^i \left(-\frac{\partial Q^i(\mathbf{x}^i)}{\partial \mathbf{x}^i} + \frac{\partial \Upsilon^i(\mathbf{x}^i)}{\partial \mathbf{x}^i} + \boldsymbol{\tau}^i \right) + \int \lambda_1^i(s) \frac{\partial \rho_i(\mathbf{x}^i, s)}{\partial \mathbf{x}^i} dF(s) = 0, \quad \forall i, \quad (\text{OA1})$$

where Equation (OA1) can be expressed as

$$-\frac{\partial Q^i(\mathbf{x}^i)}{\partial \mathbf{x}^i} + \frac{\partial \Upsilon^i(\mathbf{x}^i)}{\partial \mathbf{x}^i} + \boldsymbol{\tau}^i = \int m^i(s) \frac{\partial \rho_i(\mathbf{x}^i, s)}{\partial \mathbf{x}^i} dF(s), \quad \forall i, \quad (\text{OA2})$$

where $m^i(s) = \frac{\lambda_1^i(s)}{\lambda_0^i}$ denotes the stochastic discount factor of investor i .²⁴ These conditions are Euler equations for both financing and investment. All partial derivatives in these expressions denote vectors, where each element in the derivative corresponds to the associated element of $\mathbf{x}^i$.

Creditors' problem: The problem solved by creditors in Lagrangian form is

$$\max_{c_0^C, \{c_1^C(s)\}, \{h_i^C\}} \mathcal{L}^C,$$

²⁴Note that $\rho_i(\mathbf{x}^i, s)$ must be sufficiently well-behaved for the second term of Equation (OA1) to be valid. Otherwise, all results follow when the second term is $\frac{\partial}{\partial \mathbf{x}^i} \left[\int \lambda_1^i(s) \rho_i(\mathbf{x}^i, s) dF(s) \right]$.

where $\mathcal{L}^C$ is given by

$$\begin{aligned}\mathcal{L}^C = & u^C \left(c_0^C, \{c_1^C(s)\}_{s \in S}, \{\bar{\mathbf{x}}^j\}_{j \in \mathcal{I}} \right) - \lambda_0^C \left(c_0^C - n_0^C + \sum_{i \in \mathcal{I}} h_i^C Q^i(\bar{\mathbf{x}}^i) \right) \\ & - \int \lambda_1^C(s) \left(c_1^C(s) - n_1^C(s) - \sum_{i \in \mathcal{I}} h_i^C \rho_i^C(\bar{\mathbf{x}}^i, s) \right) dF(s),\end{aligned}$$

where λ_0^C and $\lambda_1^C(s)$ denote the Lagrange multipliers that correspond to the creditors' budget constraints. The consumption optimality conditions imply that $\lambda_0^C = \frac{\partial u^C}{\partial c_0^C}$ and $\lambda_1^C(s) dF(s) = \frac{\partial u^C}{\partial c_1^C(s)}$. The optimality conditions for creditors regarding $\{h_i^C\}$ are

$$-\lambda_0^C Q^i(\mathbf{x}^i) + \int \lambda_1^C(s) \rho_i^C(\mathbf{x}^i, s) dF(s) = 0, \forall i, \quad (\text{OA3})$$

where we use the fact that $\mathbf{x}^i = \bar{\mathbf{x}}^i$ in equilibrium. Equation (OA3) can be expressed as

$$Q^i(\mathbf{x}^i) = \int m^C(s) \rho_i^C(\mathbf{x}^i, s) dF(s), \forall i, \quad (\text{OA4})$$

where we define $m^C(s) = \frac{\lambda_1^C(s)}{\lambda_0^C}$, characterizing the financing schedules $Q^i(\mathbf{x}^i)$ that investors face. Note that the stochastic discount factor $m^C(s)$ is an equilibrium object, which depends on the choices of all investors in the model and the policy.

Proof of Lemma 1 (Marginal Welfare Effects of Regulation)

Proof. We initially characterize the $N \times 1$ vectors $\frac{dV^i}{d\tau^j}$ and $\frac{dV^C}{d\tau^j}$, which correspond to the money-metric welfare changes of type i investors and creditors when τ^j changes.

Investors: We express the financing schedules faced by investors as a function of the creditors' stochastic discount factor $m^C(s)$, which is an endogenous function of the consumption of creditors in all dates and states. This allows us to separately account for any general equilibrium pecuniary effects. Formally, we represent the equilibrium financing schedules in Equation (OA4) for an investor i as follows:

$$Q^i(\mathbf{x}^i; m^C(s)) = \int m^C(s) \rho_i^C(\mathbf{x}^i, s) dF(s),$$

where we make explicit the dependence on $m^C(s)$. The money-metric change in indirect utility for investor i when varying the regulation that investor j faces is given by the following $N \times 1$

vector:

$$\begin{aligned}
\frac{dV^i}{d\tau^j} &= \frac{dc_0^i}{d\tau^j} \overbrace{\left(\frac{\frac{\partial u^i}{\partial c_0^i} - \lambda_0^i}{\lambda_0^i} \right)}^{=0} + \int \frac{dc_1^i(s)}{d\tau^j} \overbrace{\left(\frac{\frac{\frac{\partial u^i}{\partial c_1^i(s)}}{dF(s)} - \lambda_1^i(s)}{\lambda_0^i} \right)}^{=0} dF(s) \\
&+ \frac{d\mathbf{x}^i}{d\tau^j} \overbrace{\left(\frac{\partial Q^i(\mathbf{x}^i; m^C(s))}{\partial \mathbf{x}^i} - \frac{\partial \Upsilon^i(\mathbf{x}^i)}{\partial \mathbf{x}^i} - \boldsymbol{\tau}^i + \int m^i(s) \frac{\partial \rho_i(\mathbf{x}^i, s)}{\partial \mathbf{x}^i} dF(s) \right)}^{=0} \\
&+ \frac{dT^i}{d\tau^j} - \frac{d\boldsymbol{\tau}^i}{d\tau^j} \mathbf{x}^i + \frac{\partial Q^i(\mathbf{x}^i; m^C(s))}{\partial m^C(s)} \frac{dm^C(s)}{d\tau^j} + \sum_{\ell \in \mathcal{I}} \frac{d\mathbf{x}^\ell}{d\tau^j} \frac{1}{\lambda_0^i} \frac{\partial u^i}{\partial \mathbf{x}^\ell}, \tag{OA5}
\end{aligned}$$

where the $N \times 1$ vectors $\frac{dT^i}{d\tau^j}$ and $\mathbf{x}^j$ are given by

$$\frac{dT^i}{d\tau^j} = \begin{pmatrix} \frac{dT^i}{d\tau_1^j} \\ \vdots \\ \frac{dT^i}{d\tau_N^j} \end{pmatrix} \quad \text{and} \quad \mathbf{x}^i = \begin{pmatrix} x_1^i \\ \vdots \\ x_n^i \\ \vdots \\ x_N^i \end{pmatrix},$$

and where the matrix $\frac{d\boldsymbol{\tau}^i}{d\tau^j}$, of dimension $N \times N$, is given by

$$\frac{d\boldsymbol{\tau}^i}{d\tau^j} = \begin{cases} I_N, & \text{if } i = j \\ 0_N, & \text{if } i \neq j, \end{cases}$$

which is either a N -dimensional identity matrix, I_N , when $i = j$, or a $N \times N$ matrix of zeros, 0_N , when $i \neq j$. We also define the $N \times 1$ vector $\frac{\partial Q^i(\mathbf{x}^i; m^C(s))}{\partial m^C(s)} \frac{dm^C(s)}{d\tau^j}$ as

$$\frac{\partial Q^i(\mathbf{x}^i; m^C(s))}{\partial m^C(s)} \frac{dm^C(s)}{d\tau^j} = \begin{pmatrix} \int \frac{dm^C(s)}{d\tau_1^j} \rho_i^C(\mathbf{x}^i, s) dF(s) \\ \vdots \\ \int \frac{dm^C(s)}{d\tau_N^j} \rho_i^C(\mathbf{x}^i, s) dF(s) \end{pmatrix}.$$

Note that we use the fact that

$$\frac{dQ^i(\mathbf{x}^i; m^C(s))}{d\tau^j} = \frac{d\mathbf{x}^i}{d\tau^j} \frac{\partial Q^i}{\partial \mathbf{x}^i} + \frac{\partial Q^i}{\partial m^C(s)} \frac{dm^C(s)}{d\tau^j},$$

as well as

$$\frac{d(\boldsymbol{\tau}^i \cdot \mathbf{x}^i - T^i)}{d\boldsymbol{\tau}^j} = \frac{d\boldsymbol{\tau}^i}{d\boldsymbol{\tau}^j} \mathbf{x}^i + \frac{d\mathbf{x}^i}{d\boldsymbol{\tau}^j} \boldsymbol{\tau}^i - \frac{dT^i}{d\boldsymbol{\tau}^j}.$$

Note that $\frac{\partial u^i}{\partial \bar{\mathbf{x}}^\ell}$ denotes a $N \times 1$ gradient vector.

Creditors: In the case of creditors, we can express the $N \times 1$ vector $\frac{\frac{dV^C}{d\boldsymbol{\tau}^j}}{\lambda_0^C}$ as follows:

$$\begin{aligned} \frac{\frac{dV^C}{d\boldsymbol{\tau}^j}}{\lambda_0^C} &= \frac{dc_0^C}{d\boldsymbol{\tau}^j} \overbrace{\left(\frac{\frac{\partial u^C}{\partial c_0^C} - \lambda_0^C}{\lambda_0^C} \right)}^{=0} + \int \overbrace{\left(\frac{\frac{\frac{\partial u^C}{\partial c_1^C(s)}}{dF(s)} - \lambda_1^C(s)}{\lambda_0^C} \right)}^{=0} \frac{dc_1^C(s)}{d\boldsymbol{\tau}^j} dF(s) \\ &\quad - \sum_{i \in \mathcal{I}} \frac{dh_i^C}{d\boldsymbol{\tau}^j} \overbrace{\left(Q^i(\mathbf{x}^i; m^C(s)) - \int m^C(s) \rho_i^C(\mathbf{x}^i, s) dF(s) \right)}^{=0} \\ &\quad - \sum_{i \in \mathcal{I}} h_i^C \left(\frac{dQ^i(\mathbf{x}^i; m^C(s))}{d\boldsymbol{\tau}^j} - \int m^C(s) \frac{d\rho_i^C(\mathbf{x}^i, s)}{d\boldsymbol{\tau}^j} dF(s) \right) + \sum_{\ell \in \mathcal{I}} \frac{d\mathbf{x}^\ell}{d\boldsymbol{\tau}^j} \frac{1}{\lambda_0^C} \frac{\partial u^C}{\partial \bar{\mathbf{x}}^\ell} \\ &= - \sum_{i \in \mathcal{I}} \frac{d\mathbf{x}^i}{d\boldsymbol{\tau}^j} \underbrace{\left(\frac{\partial Q^i}{\partial \mathbf{x}^i} - \int m^C(s) \frac{\partial \rho_i^C(\mathbf{x}^i, s)}{\partial \mathbf{x}^i} dF(s) \right)}_{=0} - \sum_{i \in \mathcal{I}} \frac{\partial Q^i}{\partial m^C(s)} \frac{dm^C(s)}{d\boldsymbol{\tau}^j} + \sum_{\ell \in \mathcal{I}} \frac{d\mathbf{x}^\ell}{d\boldsymbol{\tau}^j} \frac{1}{\lambda_0^C} \frac{\partial u^C}{\partial \bar{\mathbf{x}}^\ell}, \end{aligned} \tag{OA6}$$

where in the second equality we use the fact that $h_i^C = 1$ and the fact that $\frac{d\rho_i^C(\mathbf{x}^i, s)}{d\boldsymbol{\tau}^j} = \frac{d\mathbf{x}^i}{d\boldsymbol{\tau}^j} \frac{\partial \rho_i^C}{\partial \mathbf{x}^i}$, and where the $N \times 1$ vector $\frac{\partial \rho_i^C}{\partial \mathbf{x}^i}$ is given by

$$\frac{\partial \rho_i^C}{\partial \mathbf{x}^i} = \begin{pmatrix} \frac{\partial \rho_i^C}{\partial x_1^i} \\ \vdots \\ \frac{\partial \rho_i^C}{\partial x_N^i} \end{pmatrix}.$$

Note that $\frac{d\mathbf{x}^\ell}{d\boldsymbol{\tau}^j}$ is defined above and $\frac{\partial u^C}{\partial \bar{\mathbf{x}}^\ell}$ denotes a $N \times 1$ gradient vector.

Aggregation: First, we can express the sum among investors of the change in money-metric indirect utilities as follows:

$$\begin{aligned} \sum_{i \in \mathcal{I}} \frac{\frac{dV^i}{d\boldsymbol{\tau}^j}}{\lambda_0^i} &= \sum_{i \in \mathcal{I}} \left(\frac{dT^i}{d\boldsymbol{\tau}^j} - \frac{d\boldsymbol{\tau}^i}{d\boldsymbol{\tau}^j} \mathbf{x}^i \right) + \sum_{i \in \mathcal{I}} \frac{\partial Q^i}{\partial m^C(s)} \frac{dm^C(s)}{d\boldsymbol{\tau}^j} + \sum_{i \in \mathcal{I}} \sum_{\ell \in \mathcal{I}} \frac{d\mathbf{x}^\ell}{d\boldsymbol{\tau}^j} \frac{1}{\lambda_0^i} \frac{\partial u^i}{\partial \bar{\mathbf{x}}^\ell} \\ &= \sum_{i \in \mathcal{I}} \frac{d\mathbf{x}^i}{d\boldsymbol{\tau}^j} \left(\boldsymbol{\tau}^i + \sum_{\ell \in \mathcal{I}} \frac{1}{\lambda_0^\ell} \frac{\partial u^\ell}{\partial \bar{\mathbf{x}}^i} \right) + \sum_{i \in \mathcal{I}} \frac{\partial Q^i}{\partial m^C(s)} \frac{dm^C(s)}{d\boldsymbol{\tau}^j}, \end{aligned}$$

where we use the fact that

$$\sum_{i \in \mathcal{I}} \left(\frac{dT^i}{d\tau^j} - \frac{d\tau^i}{d\tau^j} x^i \right) = \sum_{i \in \mathcal{I}} \frac{dx^i}{d\tau^j} \tau^i,$$

as well as the following identity:

$$\sum_{i \in \mathcal{I}} \sum_{\ell \in \mathcal{I}} \frac{dx^\ell}{d\tau^j} \frac{1}{\lambda_0^i} \frac{\partial u^i}{\partial \bar{x}^\ell} = \sum_{i \in \mathcal{I}} \sum_{\ell \in \mathcal{I}} \frac{dx^i}{d\tau^j} \frac{1}{\lambda_0^\ell} \frac{\partial u^\ell}{\partial \bar{x}^i}.$$

Therefore, we can express $\frac{dW}{d\tau^j}$ as follows:

$$\begin{aligned} \frac{dW}{d\tau^j} &= \sum_{i \in \mathcal{I}} \frac{\frac{dV^i}{d\tau^j}}{\lambda_0^i} + \frac{\frac{dV^C}{d\tau^j}}{\lambda_0^C} \\ &= \sum_{i \in \mathcal{I}} \frac{dx^i}{d\tau^j} \left(\tau^i + \sum_{\ell \in \mathcal{I}} \frac{1}{\lambda_0^\ell} \frac{\partial u^\ell}{\partial \bar{x}^i} \right) + \sum_{i \in \mathcal{I}} \frac{dx^i}{d\tau^j} \frac{1}{\lambda_0^C} \frac{\partial u^C}{\partial \bar{x}^i} \\ &= \sum_{i \in \mathcal{I}} \frac{dx^i}{d\tau^j} \left(\tau^i - \underbrace{\left(- \left(\sum_{\ell \in \mathcal{I}} \frac{1}{\lambda_0^\ell} \frac{\partial u^\ell}{\partial \bar{x}^i} + \frac{1}{\lambda_0^C} \frac{\partial u^C}{\partial \bar{x}^i} \right) \right)}_{=\delta^i} \right). \end{aligned}$$

Hence, by switching indexes and stacking we find that

$$\frac{dW}{d\tau} = \frac{d\mathbf{x}}{d\tau} (\boldsymbol{\tau} - \boldsymbol{\delta}) = \frac{d\mathbf{x}}{d\tau} \boldsymbol{\omega},$$

as in Equation (8) in the text. □

Proof of Proposition 1. (Second-Best Policy: Perfectly Regulated Decisions)

Proof. At the second-best optimum, it must be that the marginal welfare effects of adjusting the regulation of perfectly regulated decisions (those for which the planning constraints do not bind), satisfy $\frac{dW}{d\tau^R} = 0$. Leveraging Lemma 1, we can express these optimality conditions as

$$\frac{dW}{d\tau^R} = \frac{d\mathbf{x}}{d\tau^R} (\boldsymbol{\tau} - \boldsymbol{\delta}) = \frac{d\mathbf{x}^U}{d\tau^R} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) + \frac{d\mathbf{x}^R}{d\tau^R} (\boldsymbol{\tau}^R - \boldsymbol{\delta}^R) = 0.$$

Assuming that the matrix $\frac{d\mathbf{x}^R}{d\tau^R}$ is invertible, we rearrange this expression as follows:

$$\frac{d\mathbf{x}^R}{d\tau^R} (\boldsymbol{\tau}^R - \boldsymbol{\delta}^R) = - \frac{d\mathbf{x}^U}{d\tau^R} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) \iff \boldsymbol{\tau}^R = \boldsymbol{\delta}^R - \left(\frac{d\mathbf{x}^R}{d\tau^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U),$$

which corresponds to Equation (9) in the text. $\square$

Proof of Proposition 2. (Second-Best Policy: Imperfectly Regulated Decisions)

Proof. Leveraging Lemma 1, we can express the marginal welfare effects of adjusting the regulation of imperfectly regulated decisions (those for which the planning constraints bind) as

$$\frac{dW}{d\tau^U} = \frac{d\mathbf{x}^U}{d\tau^U} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) + \frac{d\mathbf{x}^R}{d\tau^U} (\boldsymbol{\tau}^R - \boldsymbol{\delta}^R).$$

So defining the vector of Lagrange multipliers associated with the regulatory constraints by $\boldsymbol{\mu}$, the optimality conditions for such decisions are given by $\frac{dW}{d\tau^U} = \frac{d\boldsymbol{\Phi}}{d\tau^U} \boldsymbol{\mu}$, along with $\boldsymbol{\Phi}(\boldsymbol{\tau}^U) = 0$ when $\boldsymbol{\Phi}(\cdot)$ captures constraints.

From Proposition 1, we have that

$$\boldsymbol{\tau}^R - \boldsymbol{\delta}^R = \left(-\frac{d\mathbf{x}^R}{d\tau^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U),$$

so combining the last two equations we obtain

$$\begin{aligned} \frac{dW}{d\tau^U} &= \frac{d\mathbf{x}^U}{d\tau^U} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) - \frac{d\mathbf{x}^R}{d\tau^U} \left(\frac{d\mathbf{x}^R}{d\tau^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R} (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) \\ &= \frac{d\mathbf{x}^U}{d\tau^U} \left(\mathbf{I} - \underbrace{\left(\frac{d\mathbf{x}^U}{d\tau^U} \right)^{-1} \frac{d\mathbf{x}^R}{d\tau^U} \left(\frac{d\mathbf{x}^R}{d\tau^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R}}_{\equiv \mathbf{L}} \right) (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) = \frac{d\mathbf{x}^U}{d\tau^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\omega}^U, \end{aligned}$$

which corresponds to Equation (10) in the text. $\square$

C Additional Proofs and Derivations

C.1 Price Theoretic Formulation

We now show that it is possible to extend our results to a price-theoretic environment as follows. We consider an economy with a finite number $I \geq 1$ of agents (equivalently, agent types in unit measure), indexed by $i, j \in \mathcal{I}$, where $\mathcal{I} = \{1, \dots, I\}$. There are $N \geq 1$ goods (commodities) indexed by $n \in \mathcal{N}$, where $\mathcal{N} = \{1, \dots, N\}$. Agent i 's preferences are represented by

$$u^i(\mathbf{x}^i, \bar{\mathbf{x}}), \quad (\text{OA7})$$

where $\mathbf{x}^i \in \mathbb{R}_+^N$ denotes agent i 's consumption bundle, and $\bar{\mathbf{x}} = \{\bar{\mathbf{x}}^j\}_{j \in \mathcal{I}}$ denotes the collection of bundles of all agents.²⁵ Each agent takes $\bar{\mathbf{x}}$ as given, so the second argument in $u^i(\cdot)$ captures externalities across agents.

Each agent i faces a budget constraint

$$\mathbf{p} \cdot (\mathbf{x}^i - \mathbf{e}^i) + \boldsymbol{\tau}^i \cdot \mathbf{x}^i = T^i, \quad (\text{OA8})$$

where $\mathbf{e}^i \in \mathbb{R}_+^N$ denotes agent i 's endowment of goods; $\mathbf{p} \in \mathbb{R}_+^N$ is a price vector; $\boldsymbol{\tau}^i \cdot \mathbf{x}^i$ introduces a set of taxes/subsidies (regulations) specific to each agent and commodity, where $\boldsymbol{\tau}^i \in \mathbb{R}^N$; and $T^i \in \mathbb{R}$ denotes the lump-sum transfer or tax that agent i receives or faces to ensure that the planner runs a balanced budget. We denote the elements of $\mathbf{x}^i$, $\bar{\mathbf{x}}^i$, and $\boldsymbol{\tau}^i$ by x_n^i , $\bar{x}_n^i$ and τ_n^i , respectively.

For a given set of regulations $\{\boldsymbol{\tau}^i\}_{i \in \mathcal{I}}$ and transfers $\{T^i\}_{i \in \mathcal{I}}$, an *equilibrium* consists of consumption allocations $\{\mathbf{x}^i\}_{i \in \mathcal{I}}$ and a price vector $\mathbf{p}$ such that i) agents choose $\mathbf{x}^i$ to maximize utility (OA7) taking $\mathbf{p}$ and $\bar{\mathbf{x}}$ as given, subject to the budget constraint (OA8); ii) the planner's budget is balanced, so that $\sum_i T^i = \sum_i \boldsymbol{\tau}^i \cdot \mathbf{x}^i$; iii) consumption allocations are consistent in the aggregate, so $\bar{\mathbf{x}}^i = \mathbf{x}^i$, $\forall i$; and iv) markets clear, so $\sum_i \mathbf{x}^i = \sum_i \mathbf{e}^i$. We assume at all times that the model is well-behaved.

First, we characterize the change in agent i 's indirect utility, denoted by V^i , when varying a specific regulation τ_n^j . Making use of the envelope theorem, we have

$$\frac{dV^i}{d\tau_n^j} = \frac{\partial u^i}{\partial \bar{\mathbf{x}}} \cdot \frac{d\bar{\mathbf{x}}}{d\tau_n^j} - \lambda^i \left(\frac{d\mathbf{p}}{d\tau_n^j} \cdot (\mathbf{x}^i - \mathbf{e}^i) + \frac{d\boldsymbol{\tau}^i}{d\tau_n^j} \cdot \mathbf{x}^i - \frac{dT^i}{d\tau_n^j} \right),$$

where λ^i denotes the Lagrange multiplier associated with the budget constraint. We use x and

²⁵Section A of the Online Appendix includes explicit definitions of all vectors and matrices used in the paper.

$\bar{x}$ equivalently going forward, since they are equal in equilibrium.

Expressing this welfare change in money-metric terms by normalizing by agent i 's marginal value of wealth, λ^i , and aggregating across agents, we have

$$\begin{aligned}\frac{dW}{d\tau_n^j} &= \sum_i \frac{\frac{dV^i}{d\tau_n^j}}{\lambda^i} = \sum_i \frac{1}{\lambda^i} \frac{\partial u^i}{\partial \bar{x}} \cdot \frac{d\mathbf{x}}{d\tau_n^j} - \frac{d\mathbf{p}}{d\tau_n^j} \cdot \sum_i (\mathbf{x}^i - \mathbf{e}^i) - \sum_i \left(\frac{d\boldsymbol{\tau}^i}{d\tau_n^j} \cdot \mathbf{x}^i - \frac{dT^i}{d\tau_n^j} \right) \\ &= \sum_i \frac{1}{\lambda^i} \frac{\partial u^i}{\partial \bar{x}} \cdot \frac{d\mathbf{x}}{d\tau_n^j} + \sum_i \frac{d\mathbf{x}^i}{d\tau_n^j} \cdot \boldsymbol{\tau}^i,\end{aligned}$$

where the last line follows from i) market clearing, $\sum_i (\mathbf{x}^i - \mathbf{e}^i) = 0$, and ii) the fact that the planner's budget is balanced, so

$$\sum_i \frac{dT^i}{d\tau_n^j} = \sum_i \frac{d\boldsymbol{\tau}^i}{d\tau_n^j} \cdot \mathbf{x}^i + \sum_i \frac{d\mathbf{x}^i}{d\tau_n^j} \cdot \boldsymbol{\tau}^i.$$

Note that $\frac{dW}{d\tau_n^j}$ can be equivalently expressed as

$$\frac{dW}{d\tau_n^j} = \sum_i \frac{1}{\lambda^i} \sum_\ell \frac{\partial u^i}{\partial \bar{x}^\ell} \cdot \frac{d\mathbf{x}^\ell}{d\tau_n^j} + \sum_\ell \frac{d\mathbf{x}^\ell}{d\tau_n^j} \cdot \boldsymbol{\tau}^\ell = \sum_\ell \frac{d\mathbf{x}^\ell}{d\tau_n^j} (\boldsymbol{\tau}^\ell - \boldsymbol{\delta}^\ell),$$

where $\boldsymbol{\delta}^\ell = -\sum_i \frac{1}{\lambda^i} \frac{\partial u^i}{\partial \bar{x}^\ell}$, so by switching indexes and stacking we find that

$$\frac{dW}{d\boldsymbol{\tau}} = \frac{d\mathbf{x}}{d\boldsymbol{\tau}} (\boldsymbol{\tau} - \boldsymbol{\delta}) = \frac{d\mathbf{x}}{d\boldsymbol{\tau}} \boldsymbol{\omega},$$

as in Equation (8) in the text.

C.2 Game Theoretic Formulation

We now show that it is possible to extend our results to a game-theoretic environment as follows.

Suppose that agents have preferences of the form

$$u^i(\mathbf{x}^i, \bar{\mathbf{x}})$$

and that they face a constraint given by

$$\Psi^i(\mathbf{x}^i, \bar{\mathbf{x}}; \theta) = 0,$$

where θ is a parameter that indexes a general perturbation. In the price-theoretic formulation in Section C.1, the function $\Psi^i(\mathbf{x}^i, \bar{\mathbf{x}}; \theta)$ takes the simple form

$$\Psi^i(\mathbf{x}^i, \bar{\mathbf{x}}; \theta) = p(\bar{\mathbf{x}})(\mathbf{x}^i - \mathbf{e}^i) + \boldsymbol{\tau}^i(\theta) \mathbf{x}^i - T^i.$$

We consider a Walras-Nash equilibrium with taxes, in which optimality requires that $\frac{\partial u^i}{\partial \mathbf{x}^i} = \lambda^i \frac{\partial \Psi^i}{\partial \mathbf{x}^i}$. In general, we can express the marginal welfare change in the utility of agent i induced by a general perturbation as

$$\frac{dV^i}{d\theta} = \frac{\partial u^i}{\partial \mathbf{x}^i} \cdot \frac{d\mathbf{x}^i}{d\theta} + \frac{\partial u^i}{\partial \bar{\mathbf{x}}} \cdot \frac{d\bar{\mathbf{x}}}{d\theta} - \lambda^i \left(\frac{\partial \Psi^i}{\partial \mathbf{x}^i} \cdot \frac{d\mathbf{x}^i}{d\theta} + \frac{\partial \Psi^i}{\partial \bar{\mathbf{x}}} \cdot \frac{d\bar{\mathbf{x}}}{d\theta} + \frac{\partial \Psi^i}{\partial \theta} \right),$$

and normalizing by λ^i to express the marginal welfare change in units of the constraint

$$\frac{dV^i}{\lambda^i} = \left(\frac{1}{\lambda^i} \frac{\partial u^i}{\partial \bar{\mathbf{x}}} - \frac{\partial \Psi^i}{\partial \bar{\mathbf{x}}} \right) \cdot \frac{d\bar{\mathbf{x}}}{d\theta} - \frac{\partial \Psi^i}{\partial \theta}.$$

This normalization allows us to express aggregate welfare gains as

$$\frac{dW}{d\theta} = \sum_i \frac{dV^i}{\lambda^i} = \sum_i \left(\frac{1}{\lambda^i} \frac{\partial u^i}{\partial \bar{\mathbf{x}}} - \frac{\partial \Psi^i}{\partial \bar{\mathbf{x}}} \right) \cdot \frac{d\bar{\mathbf{x}}}{d\theta} - \sum_i \frac{\partial \Psi^i}{\partial \theta}. \quad (\text{OA9})$$

This expression is the generalization of Equation (8) in Lemma 1, where the term multiplying $\frac{d\bar{\mathbf{x}}}{d\theta}$ exactly defines marginal distortions, and the term $\sum_i \frac{\partial \Psi^i}{\partial \theta}$ captures the direct impact of any policy perturbation. When the policy perturbation takes the form $\boldsymbol{\tau}^i(\theta) \mathbf{x}^i$, then Equation (OA9) is exactly a generalized version of Lemma 1, with a slightly redefined marginal distortion, which is now augmented to capture interactions among agents via constraints.

C.3 Redistributive Concerns

Given the money-metric marginal welfare effects of varying $\boldsymbol{\tau}^j$, defined in Equations (OA5) and (OA6), we can express the marginal welfare effects of varying $\boldsymbol{\tau}^j$, for any set of generalized social marginal welfare weights (Saez and Stantcheva, 2016) or individual weights (Dávila and Schaab, 2025), as follows:

$$\frac{dW}{d\boldsymbol{\tau}^j} = \sum_i \omega^i \frac{dV^i}{d\boldsymbol{\tau}^j} + \omega^C \frac{dV^C}{d\boldsymbol{\tau}^j} = \underbrace{\sum_i \frac{dV^i}{d\boldsymbol{\tau}^j} \frac{1}{\lambda_0^i}}_{\text{Efficiency}} + \underbrace{\frac{dV^C}{d\boldsymbol{\tau}^j} \frac{1}{\lambda_0^C} + \text{Cov}_{iC}^\Sigma \left[\omega^{iC}, \frac{dV^{iC}}{d\boldsymbol{\tau}^j} \right]}_{\text{Redistribution}}, \quad (\text{OA10})$$

where we assume, without loss of generality, that the weights average to one, that is, $\frac{1}{I} \sum_i \omega^i + \omega^C = 1$, and where we use the index iC to refer to the set of investors and creditors.²⁶

When $\omega^i = \omega^C = 1$, then the redistribution term in Equation (OA10) is zero. This case is the one studied in the body of the paper. When $\omega^i \neq \omega^C \neq 1$, Equation (OA10) shows that redistributive concerns enter additively to the marginal welfare effects of varying τ^j . A utilitarian planner simply corresponds to setting marginal welfare weights of the form $\omega^i = \lambda_0^i$, where λ_0^i typically equals marginal utility of consumption. Note that a utilitarian planner with access to lump-sum taxes/transfers finds it optimal to endogenously set $\omega^i = \omega^C = 1$.

C.4 Practical Scenarios

C.4.1 Uniform Regulation

To build intuition for the uniform regulation results, it is useful to first consider the special case where *all* decisions are subject to uniform regulation ($\mathbf{x}^U = \mathbf{x}$). In that case, it follows from Proposition 2 that the optimal uniform regulation is given by

$$\bar{\tau}^U = \frac{\sum_i \sum_n \frac{dx_n^i}{d\bar{\tau}^U} \delta_n^i}{\sum_j \sum_n \frac{dx_n^j}{d\bar{\tau}^U}} = \frac{\sum_i \sum_n w_n^i \delta_n^i}{\sum_j \sum_n \frac{dx_n^j}{d\bar{\tau}^U}}, \quad (\text{OA11})$$

where we have re-written the total response of decision x_n^j to the uniform regulation as

$$\frac{dx_n^j}{d\bar{\tau}^U} = \sum_{j' \in \mathcal{I}} \sum_{n' \in \mathcal{N}} \frac{dx_n^j}{d\tau_{n'}^{j'}}.$$

In general, note that the case of uniform regulation is a particular case of linear constraints $\mathbf{A}\boldsymbol{\tau}^U = 0$, where

$$\mathbf{A} = \begin{pmatrix} 1 & -1 & & \cdots & 0 \\ & 1 & -1 & & \vdots \\ \vdots & & \ddots & \ddots & \\ 0 & \cdots & & 1 & -1 \end{pmatrix},$$

where $\mathbf{A}$ has dimensions $(N_U - 1) \times N_U$. In this case, $\frac{d\Phi}{d\boldsymbol{\tau}} = \mathbf{A}'$, so Proposition 2 implies that the regulator optimally sets

$$\frac{dW}{d\boldsymbol{\tau}^U} = \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\omega}^U = \mathbf{A}' \boldsymbol{\mu},$$

²⁶In Equation (OA10), the covariance operator, indexed by iC , corresponds to a cross-sectional covariance-sum including investors and creditors.

which also implies that

$$\boldsymbol{\iota}' \frac{dW}{d\boldsymbol{\tau}^U} = \boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\omega}^U = \boldsymbol{\iota}' \mathbf{A}' \boldsymbol{\mu} = 0,$$

since $\mathbf{A}\boldsymbol{\iota} = 0$, where $\boldsymbol{\iota}$ denotes a vector of ones. Rearranging, we obtain

$$\begin{aligned} 0 &= \boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\omega}^U \\ &= \boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) \\ &= \boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) (\bar{\tau}^U \boldsymbol{\iota} - \boldsymbol{\delta}^U), \end{aligned}$$

where the last line uses the fact that all elements of $\boldsymbol{\tau}^U$ must be equal to the same scalar, denoted $\bar{\tau}^U$, at the constrained solution. We solve as follows for the scalar $\bar{\tau}^U$ to complete the derivation of Equation (11):

$$\underbrace{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\iota} \bar{\tau}^U}_{\text{scalar}} = \underbrace{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\delta}^U}_{\text{scalar}} \iff \bar{\tau}^U = \frac{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\delta}^U}{\boldsymbol{\iota}' \frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) \boldsymbol{\iota}}.$$

C.4.2 Convex Costs of Regulation

In this case, we have $\frac{d\Phi}{d\boldsymbol{\tau}} = \mathbf{B}\boldsymbol{\tau}^U$, so Proposition 2 implies that the planner optimally sets

$$\frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} (\mathbf{I} - \mathbf{L}) (\boldsymbol{\tau}^U - \boldsymbol{\delta}^U) = \mathbf{B}\boldsymbol{\tau}^U.$$

Solving for $\boldsymbol{\tau}^U$ yields

$$\boldsymbol{\tau}^U = \left(\mathbf{B} + \left(-\frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} \right) (\mathbf{I} - \mathbf{L}) \right)^{-1} \left(\left(-\frac{d\mathbf{x}^U}{d\boldsymbol{\tau}^U} \right) (\mathbf{I} - \mathbf{L}) \boldsymbol{\delta}^U \right), \quad (\text{OA12})$$

which establishes Equation (12). In general, the correction relative to the first-best policy is given by $(\mathbf{B} + \mathbf{K})^{-1} \mathbf{K}$, which has the interpretation of an attenuation matrix. For instance, in a scenario with a single agent, $I = 1$, and a single decision, $N = 1$, Equation (OA12) becomes

$$\tau^U = \frac{\left(-\frac{dx^U}{d\tau^U} \right)}{b + \left(-\frac{dx^U}{d\tau^U} \right)} \delta^U, \quad (\text{OA13})$$

where b is a non-negative scalar that modulates the cost. In the well-behaved case in which $\frac{dx}{d\tau} < 0$, it follows that the optimal regulation is simply a scaled-down version of the first-best

regulation.²⁷

D Direct Measurement: Additional Results

Here we provide additional details on the direct measurement results in Section 4.

D.1 Empirical Estimates of Leakage Elasticities

This appendix explains our measurement of leakage elasticities, reported in Table 1 in the paper. Our estimates are based on Buchak, Matvos, Piskorski, and Seru (2018). Their estimates use geographic variation in changes in banks' regulatory capital ratios between 2008 and 2015. Increases in equity/asset ratios over this period capture stricter bank regulation mandated in the Dodd-Frank Act. County-level variation is driven by differences in the balance sheets of traditional banks across regions before Dodd-Frank. As an instrumental variable for these differences, Buchak, Matvos, Piskorski, and Seru (2018) use changes in banks' capital ratios during the 2007-8 financial crisis. The instrument captures plausibly exogenous variation in exposure to the crisis, with greater exposure leading to a larger additional regulatory burden under Dodd-Frank.

Buchak, Matvos, Piskorski, and Seru (2018) present the following two regressions to assess the effect of changes in bank capital ratios $\Delta\text{Capital ratio}_c$ across counties, indexed by c :

$$\begin{aligned}\Delta\text{Shadow bank share}_c &= \beta_0^{SS} + \beta_1^{SS}\Delta\text{Capital ratio}_c + X_c'\Gamma^{SS} + \epsilon_c \\ \Delta\text{Shadow bank lending}_c &= \beta_0^{SL} + \beta_1^{SL}\Delta\text{Capital ratio}_c + X_c'\Gamma^{SL} + \epsilon_c,\end{aligned}$$

where X_c is a vector of county-level controls, and where capital ratios are scaled by the standard deviation across counties, which is $\sigma = 0.04135$ (supplied to us by the authors). The outcome variables are defined as follows:

$$\Delta\text{Shadow bank share}_c = 100 \times \left(\frac{\text{Shadow bank originations}_{c,2015}}{\text{All originations}_{c,2015}} - \frac{\text{Shadow bank originations}_{c,2008}}{\text{All originations}_{c,2008}} \right)$$

²⁷In this case, note that the perfectly regulated decisions in turn must satisfy:

$$\tau^R = \delta^R + \left(-\frac{d\mathbf{x}^R}{d\tau^R} \right)^{-1} \frac{d\mathbf{x}^U}{d\tau^R} \left((\mathbf{B} + \mathbf{K})^{-1} \mathbf{K} - \mathbf{I} \right) \delta^U.$$

and

$$\Delta \text{Shadow bank lending}_c = 100 \times \left(\frac{\text{Shadow bank originations}_{c,2015} - \text{Shadow bank originations}_{c,2008}}{\text{All originations}_{c,2015}} \right).$$

We can map the coefficients of these regressions to the variables in our model:

$$\beta_1^{SS} = 100\sigma \times \frac{d\varphi^U}{d\tau_b^R} = 100\sigma \times \varphi^U (1 - \varphi^U) \left(\frac{d \log k_0^U}{d\tau_b^R} - \frac{d \log k_0^R}{d\tau_b^R} \right),$$

where $\varphi^U = \frac{k_0^U}{k_0^U + k_0^R}$ is the market share of shadow banks, and

$$\beta_1^{SL} = 100\sigma \times \frac{dk_0^U}{d\tau_b^R} \frac{1}{k_0^U + k_0^R} = 100\sigma \times \varphi^U \frac{d \log k_0^U}{d\tau_b^R}.$$

Using these expressions, we obtain formulae for the leakage elasticities $\frac{d \log k_0^U}{d\tau_b^R}$ and $\frac{d \log k_0^R}{d\tau_b^R}$ in terms of reduced-form estimates β_1^{SL} and β_1^{SS} and the shadow bank market share φ^U . First, we have

$$\frac{d \log k_0^U}{d\tau_b^R} = \frac{1}{100\sigma} \frac{\beta_1^{SL}}{\varphi^U}. \quad (\text{OA14})$$

Second, we have

$$\frac{1}{100\sigma} \frac{\beta_1^{SS}}{\varphi^U (1 - \varphi^U)} = \frac{d \log k_0^U}{d\tau_b^R} - \frac{d \log k_0^R}{d\tau_b^R} = \frac{1}{100\sigma} \frac{\beta_1^{SL}}{\varphi^U} - \frac{d \log k_0^R}{d\tau_b^R},$$

which we rearrange to obtain

$$\frac{d \log k_0^R}{d\tau_b^R} = \frac{1}{100\sigma} \left(\frac{\beta_1^{SL}}{\varphi^U} - \frac{\beta_1^{SS}}{\varphi^U (1 - \varphi^U)} \right). \quad (\text{OA15})$$

We derive the estimates for leakage reported in Table 1 by using a shadow bank share of $\varphi^U = 0.5$, consistent with the data reported in [Buchak, Matvos, Piskorski, and Seru \(2018\)](#) for 2015, as well as the IV coefficient estimates reported by [Buchak, Matvos, Piskorski, and Seru \(2018, Table 8B\)](#), which are

$$\hat{\beta}_1^{SL} = 8.241 \quad \text{and} \quad \hat{\beta}_1^{SS} = 9.634.$$

Substituting into (OA14) and (OA15), we obtain the calibration in Table 1:

$$\begin{aligned} \frac{d \log k_0^U}{d\tau_b^R} &= \frac{1}{100\sigma} \frac{8.241}{0.5} = 3.9860 \\ \frac{d \log k_0^R}{d\tau_b^R} &= \frac{1}{100\sigma} \left(\frac{8.241}{0.5} - \frac{9.634}{0.25} \right) = -5.3335. \end{aligned}$$

D.2 Supplementary Quantitative Results

Figures OA-1 and OA-2 present additional quantitative results using the empirical estimates outlined in Section 4.1. Figure OA-1 extends the results presented in Figure 1 in the paper to a finer grid of values for the distortion λ generated by shadow banks relative to traditional banks. These results are consistent with the discussion in the text, since welfare effects are linear in λ , and are presented for completeness.

Figure OA-2 repeats the analysis in Figure OA-1 for values of the leverage leakage elasticity $\frac{db^U}{d\tau_b^R} \neq 0$. In this case, we use the parametrization $\xi = -\frac{\omega_b^U \frac{db^U}{d\tau_b^R}}{\omega_b^R \frac{db^R}{d\tau_b^R}}$, so that ξ measures the fraction of the direct welfare benefit that is offset by leakage to shadow bank leverage. For the range of values for ξ up to 60% that we consider, the presence of leverage leakage does not substantially change the conclusions we draw in the text.

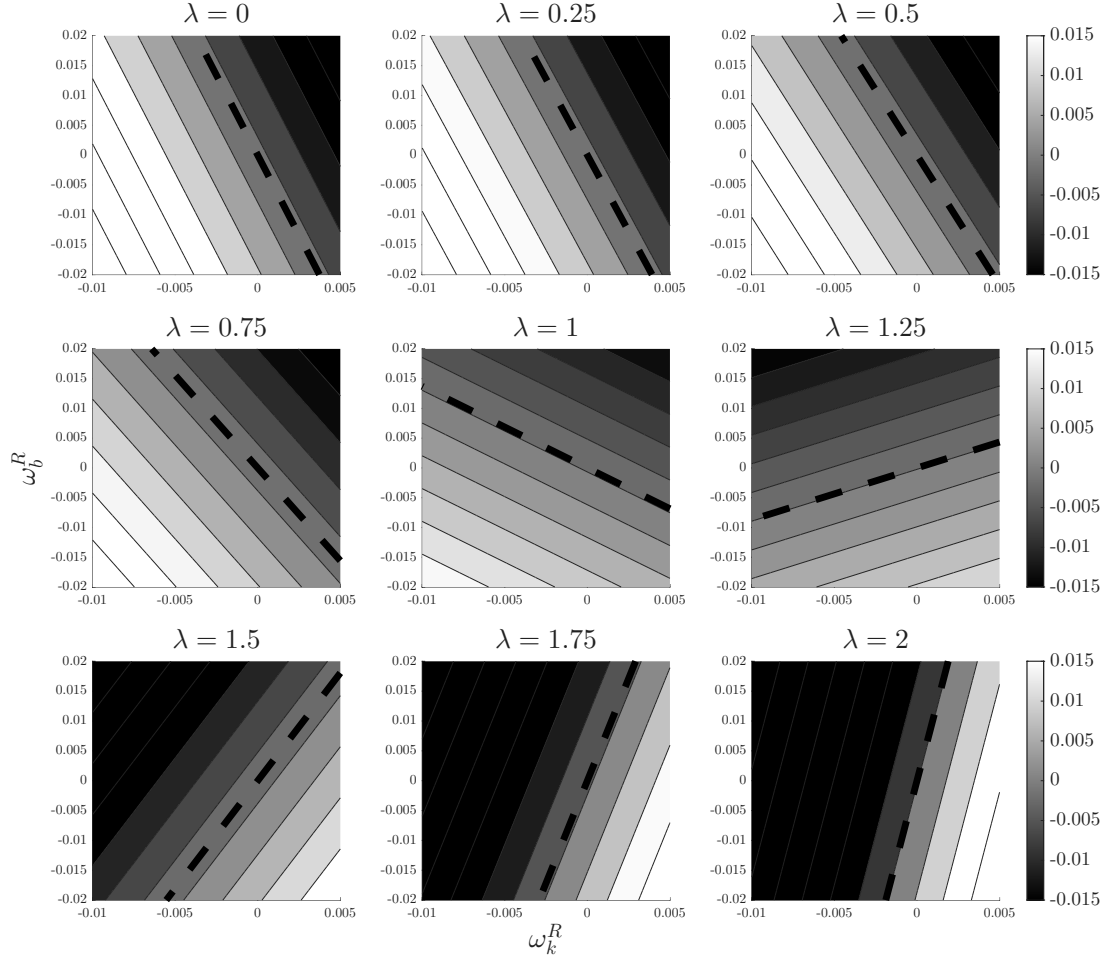


Figure OA-1: Marginal Welfare Effects with Finer Grid for λ

Note: This figure plots $\frac{dW}{d\tau_b^R}$, the marginal welfare effect of tighter leverage regulation, as a function of Pigouvian wedges $\{\omega_k^R, \omega_k^U, \omega_b^R\}$ using the empirical estimates for leakage elasticities from Table 1, setting $\omega_k^U = \lambda \omega_k^R$. The thick dashed lines show combinations of wedges for which $\frac{dW}{d\tau_b^R} = 0$. The figure is equivalent to Figure 1 in the text, but uses a finer grid of values for the relative externality parameter λ .

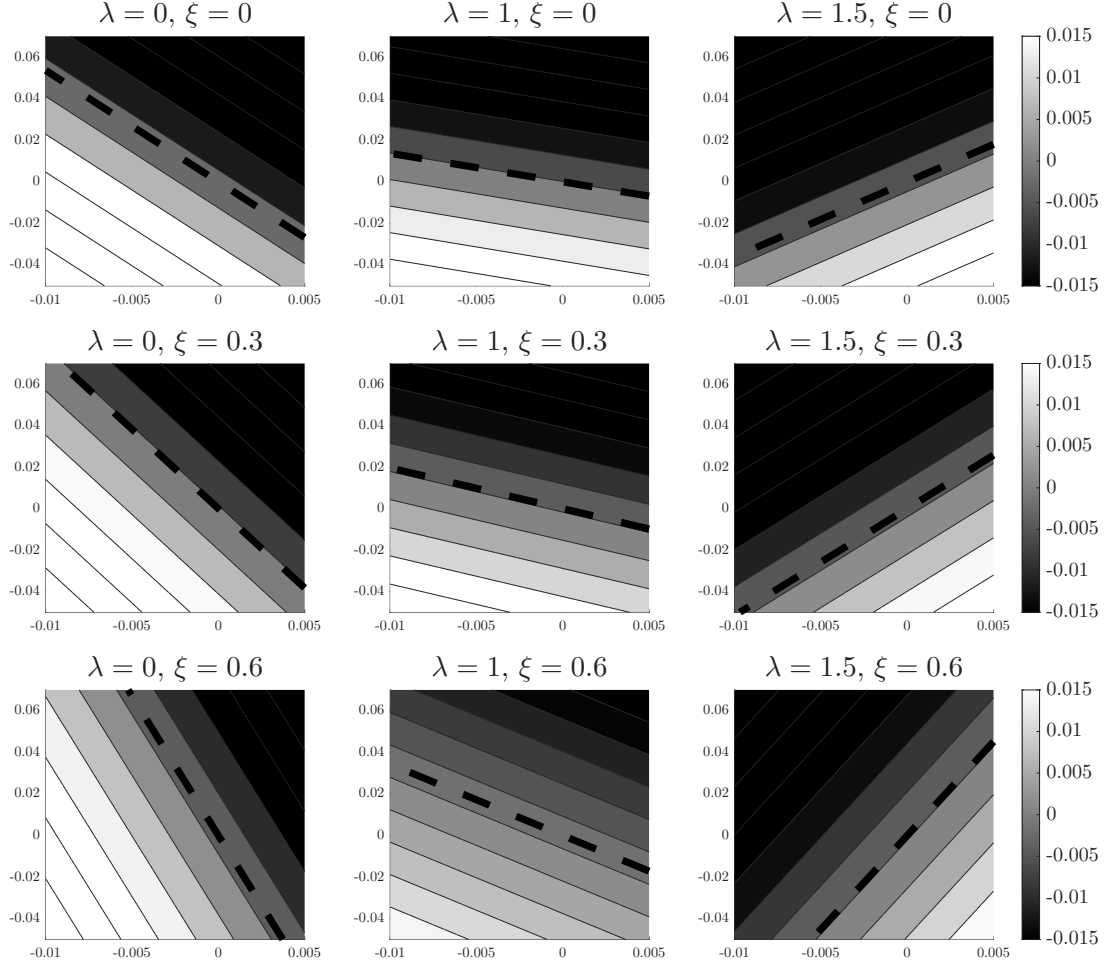


Figure OA-2: Marginal Welfare Effects with Leakage $\frac{db^U}{d\tau_b^R} \neq 0$

Note: This figure plots $\frac{dW}{d\tau_b^R}$, the marginal welfare effect of tighter leverage regulation, as a function of Pigouvian wedges $\{\omega_k^R, \omega_k^U, \omega_b^R, \omega_b^U\}$, in cases where $\frac{db^U}{d\tau_b^R} \neq 0$. We again use the empirical estimates for leakage elasticities from Table 1, setting $\omega_k^U = \lambda \omega_k^R$, and $\xi = -\frac{\omega_b^U \frac{db^U}{d\tau_b^R}}{\omega_b^R \frac{db^R}{d\tau_b^R}}$. The thick dashed lines show combinations of wedges for which $\frac{dW}{d\tau_b^R} = 0$. The figure is equivalent to Figure 1 in the text, but relaxes our baseline assumption that leakage from traditional bank regulation to shadow bank leverage is close to zero.

E Model Quantification: Additional Results

In this section, we provide detailed derivations of how the equilibrium and the normative results of the model in the quantitative application in Section 4.2 are characterized and computed. We work with more general notation: We write the potential payoffs from investment to investor type i as v^i , and the realized payoffs as $\rho_1^i(z)$ at date 1, where z is the (discrete) realization of an interim state, and $\rho_2^i(s, z)$ at date 2, where s is the realization of the final (continuous) state. We also allow agents to disagree on the distribution of all states: We denote the subjective probability that agent $i \in \{R, U, C, O\}$ attaches to state z by $\pi^i(z)$, and the subjective cumulative distribution of s conditional on observing z by $F^i(s|z)$. The planner computes welfare using $\pi^P(z)$ and $F^P(s|z)$ for all agents. To recover the quantitative model in the paper, set $\rho_1^i(z) = (1 - z)\eta^i v^i$, and $\rho_2^i(z, s) = (1 - \eta^i)v^i + zs$, and assume that $F^i(s|z) = F^P(s|z) = F(s)$ for all i .

E.1 Equilibrium Characterization: Detailed Derivations

Agents' utility, given risk-neutrality, is equal to:

$$c_0^i + \beta^i \sum_z \pi^i(z) \left(c_1^i(z) + \int c_2^i(z, s) dF^i(s|z) \right).$$

where $c_t(\cdot)$ denotes state-contingent consumption. We assume that creditors' and outsiders' endowments are sufficiently large to ensure that their non-negativity constraints on consumption never bind. As in the paper, we characterize the solution of the model by backward induction, starting from date 2. Investors' budget constraints at date 0 are:

$$c_0^i = n_0^i + Q^i(b^i) k_0^i - p_0 k_0^i - \Psi^i(k_0^i) - \tau_b^i b^i k_0^i - \hat{\tau}_k^i k_0^i + T_0^i. \quad (\text{OA16})$$

At date 1 and 2, investors' budget constraints are:

$$c_1^i(z) = \left(\rho_1^i(z) - \zeta^i b^i \right) k_0^i + p_1(z) \left(k_0^i - k_1^i(z) \right), \quad (\text{OA17})$$

and

$$c_2^i(z, s) = \begin{cases} \rho_2^i(s, z) k_1^i(z) - ((1 - \zeta^i) b^i - t^i(b^i, s, z)) k_0^i, & (z, s) \in \mathcal{N}^i \\ 0, & (z, s) \in \mathcal{D}^i, \end{cases} \quad (\text{OA18})$$

where $\mathcal{N}^i$ and $\mathcal{D}^i$ respectively denote the no-default and default regions, which are determined in equilibrium.

Creditors' budget constraints are

$$c_0^C = n_0^C - \sum_i h_i^C Q^i(b^i) k_0^i \quad \text{and} \quad c_1^C(z) = \sum_i h_i^C \zeta^i b^i k_0^i, \quad (\text{OA19})$$

at date 0, and

$$c_2^C(z, s) = \sum_i h_i^C \mathcal{R}^i(z, s) k_0^i - (1 + \kappa) \sum_i t^i(b^i, s, z) k_0^i, \quad (\text{OA20})$$

at date 2, where $\mathcal{R}^i(z, s)$ denotes the realized debt repayment to investors, which is given — depending on whether or not investors default — by

$$\mathcal{R}^i(z, s) = \begin{cases} (1 - \zeta^i) b^i & (z, s) \in \mathcal{N}^i \\ \phi \rho_2^i(s, z) k_1^i(z) / k_0^i + t^i(b^i, s, z), & (z, s) \in \mathcal{D}^i. \end{cases} \quad (\text{OA21})$$

Finally, outsiders' budget constraints are given by

$$c_0^O = p_0 k_0^O - \Upsilon(k_0^O) \quad \text{and} \quad c_1^O(z) = H(k_1^O(z)) - p_1(z) k_1^O(z). \quad (\text{OA22})$$

E.1.1 Default Decision (Date 2)

At date 2, after s is realized (z was realized at date 1), banks decide whether to default or not. It is optimal to default when

$$\rho_2^i(s, z) k_1^i(z) - \left((1 - \zeta^i) b^i - t^i(b^i, s, z) \right) k_0^i < 0. \quad (\text{OA23})$$

Hence, provided that the left-hand side of Equation (OA23) is increasing in s , there exists a unique default threshold for each bank type, $\hat{s}^i(z)$, which solves:

$$\rho_2^i(\hat{s}^i(z), z) k_1^i(z) = \left((1 - \zeta^i) b^i - t^i(b^i, \hat{s}^i(z), z) \right) k_0^i. \quad (\text{OA24})$$

Investors do not default when $\hat{s}^i(z) < \underline{s}$, an event that we denote by $\mathcal{N}^i$, and default when $\hat{s}^i(z) > \underline{s}$, denoted $\mathcal{D}^i$.

E.1.2 Investment Sale Decision (Date 1)

At date 1, once z is realized, banks choose $k_1^i(z)$, given b^i and k_0^i , determined at date 0, and given a price of investments $p_1(z)$, determined in equilibrium. The relevant continuation objective for

banks at that point is given by

$$c_1^i(z) + \int c_2^i(z, s) dF^i(s|z) = \left(\int_{\bar{s}^i(z; k_1^i(z))}^{\bar{s}} \rho_2^i(s, z) dF^i(s|z) - p_1(z) \right) k_1^i(z) + \dots$$

Hence, under the sustained assumption that date 2 payoffs are sufficiently desirable, so $\int_{\bar{s}^i(z; k_1^i(z))}^{\bar{s}} \rho_2^i(s, z) dF^i(s|z) > p_1(z)$, banks optimally choose $k_1^{i*}(z) = k_0^i$, provided that $c_1^i(z) \geq 0$.

There are two possibilities. First, if $\rho_1^i(z) < \zeta^i b^i$, then $c_1^i(z) = 0$ is optimal and a fire-sale event ensues. In this case, Equation (OA17) implies that

$$k_1^{i*}(z) = \left(1 - \frac{\zeta^i b^i - \rho_1^i(z)}{p_1(z)} \right) k_0^i \iff k_0^i - k_1^{i*}(z) = \frac{\zeta^i b^i - \rho_1^i(z)}{p_1(z)} k_0^i.$$

Second, if $\rho_1^i(z) > \zeta^i b^i$, then $c_1^i(z) > 0$ and $k_1^{i*}(z) = k_0^i$, with date 1 consumption given by

$$c_1(z) = \left(\rho_1^i(z) - \zeta^i b^i \right) k_0^i > 0.$$

Therefore, we can express banks' date 1 decision as

$$k_1^{i*}(z) = \left(1 - \frac{\max\{\zeta^i b^i - \rho_1^i(z), 0\}}{p_1(z)} \right) k_0^i \iff k_0^i - k_1^{i*}(z) = \frac{\max\{\zeta^i b^i - \rho_1^i(z), 0\}}{p_1(z)} k_0^i.$$

E.1.3 Equilibrium Price (Date 1)

Outsiders' date 1 optimality condition is given by $p_1(z) = H'(k_1^O(z))$. When combined with market clearing, we can express the date 1 equilibrium price $p_1^*(z)$ as the following fixed point:

$$p_1^*(z) = H' \left(\sum_{i \in \{R, U\}} (k_0^i - k_1^{i*}(z)) \right) = H' \left(\frac{\sum_{i \in \{R, U\}} \max\{\zeta^i b^i - \rho_1^i(z), 0\} k_0^i}{p_1^*(z)} \right) \Rightarrow p_1^*(z; \mathbf{x}), \quad (\text{OA25})$$

where the notation $p_1^*(z; \mathbf{x})$ collects date 0 investment and leverage decisions in a set of “big-K” decisions $\mathbf{x} = \{\{k_0^i\}_i, \{b^i\}_i\}$ — see e.g. [Dávila and Korinek \(2018\)](#) in a similar context. Hence, the function $p_1^*(z; \mathbf{x})$ determines the equilibrium price of investments at date 1 in state z for given date 0 investment and leverage decisions of banks.

Given our functional form assumption, $H(k_1^O) = \frac{(k_1^O + 1)^\gamma}{\gamma}$, we have that $H'(k_1^O) = (k_1^O + 1)^{\gamma-1}$, so $p_1^*(z; \mathbf{x})$ takes the form

$$p_1^*(z; \mathbf{x}) = \left(\frac{\sum_i \max\{\zeta^i b^i - \rho_1^i(z), 0\} k_0^i}{p_1^*(z; \mathbf{x})} + 1 \right)^{\gamma-1},$$

which yields a unique solution when $\gamma < 1$.

E.1.4 Banks' Continuation Indirect Utility (Date 1)

At this stage, it is convenient to define equilibrium investment sales at date 1 and default at date 2 as functions of $\mathbf{x} = \{\{k_0^i\}_i, \{b^i\}_i\}$. We define i) the equilibrium share of banks' retained investments, $\theta^i(b^i, z; \mathbf{x}) \in [0, 1]$, as

$$\theta^i(b^i, z; \mathbf{x}) = \frac{k_1^{i*}(z)}{k_0^i} = 1 - \frac{\max\{\zeta^i b^i - \rho_1^i(z), 0\}}{p_1^*(z; \mathbf{x})},$$

and ii) the equilibrium default threshold $\hat{s}^i(b^i, z; \mathbf{x})$ as the implicit solution to Equation (OA24), into which we substitute $k_1^{i*}(z) = k_0^i \theta^i(b^i, z; \mathbf{x})$ to make explicit how the default threshold depends on b^i and $\mathbf{x}$. Note that $1 - \theta^i(b^i, z; \mathbf{x}) = \frac{k_0^i - k_1^{i*}(z)}{k_0^i}$, which in turn implies that $k_0^i - k_1^{i*}(z) = (1 - \theta^i(b^i, z; \mathbf{x})) k_0^i$. Note also that even though we express θ^i and $\hat{s}^i$ in terms of $\mathbf{x}$, we will initially take derivatives in terms of $p_1^*(z; \mathbf{x})$.

Therefore, we can express the continuation indirect utility of type i banks from date 1 onwards as

$$\nu_1^i(k_0^i, b^i, z; \mathbf{x}) = c_1^i(z) + \int_{\hat{s}^i(b^i, z; \mathbf{x})}^{\bar{s}} c_2^i(z, s) dF^i(s|z) = e^i(b^i, z; \mathbf{x}) k_0^i,$$

where $e^i(b^i, z; \mathbf{x})$ denotes the payoff per unit of date 0 investment, given by

$$e^i(b^i, z; \mathbf{x}) = \max\{\rho_1^i(z) - \zeta^i b^i, 0\} + \int_{\hat{s}^i(b^i, z; \mathbf{x})}^{\bar{s}} \left(\rho_2^i(s, z) \theta^i(b^i, z; \mathbf{x}) - \left((1 - \zeta^i) b^i - t^i(b^i, s, z) \right) \right) dF^i(s|z). \quad (\text{OA26})$$

E.1.5 Debt Pricing Schedule/Credit Surface (Date 0)

At date 0, accounting for banks' default and investment sale decisions at date 1, which we have already characterized, competitive creditors price debt according to

$$Q^i(b^i; \mathbf{x}) = \beta^C \left(\zeta^i b^i + \mathbb{E}^C [\mathcal{R}^i(z, s)] \right), \quad (\text{OA27})$$

where the expected date 2 repayment (including bailout transfers) is given by

$$\begin{aligned} \mathbb{E}^C [\mathcal{R}^i(z, s)] &= \sum_z \pi^C(z) \int \mathcal{R}^i(z, s) dF^{i,C}(s|z) \\ &= \sum_z \pi^C(z) \left(\int_{\underline{s}}^{\hat{s}^i(b^i, z; \mathbf{x})} (\phi \rho_2^i(s, z) \theta^i(b^i, z; \mathbf{x}) + t^i(b^i, s, z)) dF^{i,C}(s|z) \right. \\ &\quad \left. + (1 - \zeta^i) b^i \int_{\hat{s}^i(b^i, z; \mathbf{x})}^{\bar{s}} dF^{i,C}(s|z) \right). \end{aligned}$$

We index $F^{i,C}$ (and later $F^{i,C,P}$) by i to account for the fact that $\eta^R \neq \eta^U$.

E.1.6 Banks' Problem (Date 0)

At date 0, since $c_0^i > 0$, which we always assume, banks maximize

$$c_0^i + \beta \sum_z \pi(z) e^i(b^i, z; \mathbf{x}) k_0^i = \left(M^i(b^i; \mathbf{x}) - p_0 \right) k_0^i - \Psi^i(k_0^i) - \tau_b^i b^i k_0^i - \hat{\tau}_k^i k_0^i, \quad (\text{OA28})$$

where we define the date 0 value of investment as the sum of payoffs that go to banks, $\mathcal{E}^i(b^i; \mathbf{x})$, and creditors, $Q^i(b^i; \mathbf{x})$, as follows:

$$M^i(b^i; \mathbf{x}) = \mathcal{E}^i(b^i; \mathbf{x}) + Q^i(b^i; \mathbf{x}), \quad \text{where} \quad \mathcal{E}^i(b^i; \mathbf{x}) = \beta \sum_z \pi^i(z) e^i(b^i, z; \mathbf{x}), \quad (\text{OA29})$$

where $e^i(b^i, z; \mathbf{x})$ is defined in (OA26), and where $Q^i(b^i; \mathbf{x})$ is defined in (OA27).

Leverage Decision. Given the banks' date 0 objective in (OA28), their optimal leverage decision is made independently of k_0^i , as follows:

$$\frac{\partial M^i(b^i; \mathbf{x})}{\partial b^i} = \frac{\partial \mathcal{E}^i(b^i; \mathbf{x})}{\partial b^i} + \frac{\partial Q^i(b^i; \mathbf{x})}{\partial b^i} = \tau_b^i, \quad (\text{OA30})$$

where $\frac{\partial \mathcal{E}^i(b^i; \mathbf{x})}{\partial b^i}$ and $\frac{\partial Q^i(b^i; \mathbf{x})}{\partial b^i}$ are characterized below. Note that our assumptions ensure that banks' leverage is independent of their investment decision, which simplifies the computation of the model.

Investment Decision. Given their optimal leverage decision, banks optimally choose their date 0 investment holdings as follows:

$$M^i(b^i; \mathbf{x}) = p_0 + \Psi^{i'}(k_0^i) + \tau_k^i, \quad (\text{OA31})$$

where $\tau_k^i = \hat{\tau}_k^i + \tau_b^i b^i$. Given our functional form assumption, $\Psi^i(k_0^i) = \psi^i \frac{(k_0^i)^{\nu^i}}{\nu^i}$, banks' date 0 investment demand can be expressed as

$$k_0^i = \left(\frac{1}{\psi^i} \left(M^i(b^i; \mathbf{x}) - p_0 - \tau_k^i \right) \right)^{\frac{1}{\nu^i - 1}}. \quad (\text{OA32})$$

Continuation Indirect Utility Derivatives. In order to characterize $\frac{\partial \mathcal{E}^i(b^i; \mathbf{x})}{\partial b^i}$ and $\frac{\partial Q^i(b^i; \mathbf{x})}{\partial b^i}$, it is useful to first compute $\frac{\partial \theta^i(b^i, z; \mathbf{x})}{\partial b^i}$ and $\frac{\partial \hat{s}^i(b^i, z; \mathbf{x})}{\partial b^i}$, which are given by

$$\frac{\partial \theta^i(b^i, z; \mathbf{x})}{\partial b^i} = \begin{cases} 0, & \text{if } z = 0 \\ -\frac{\zeta^i}{p_1^*(z; \mathbf{x})}, & \text{if } z = 1, \end{cases}$$

and

$$\frac{\partial \hat{s}^i(b^i, z; \mathbf{x})}{\partial b^i} = \begin{cases} \frac{1 - \zeta^i - \alpha_b^i(z)}{\theta^i(b^i, z; \mathbf{x})} & \text{if } z = 0, \\ \frac{1 - \zeta^i - \alpha_b^i(z)}{\theta^i(b^i, z; \mathbf{x})} - \frac{\hat{s}^i(b^i, z; \mathbf{x})}{\theta^i(b^i, z; \mathbf{x})} \frac{\partial \theta^i(b^i, z; \mathbf{x})}{\partial b^i}, & \text{if } z = 1. \end{cases}$$

Hence, we can compute $\frac{\partial \mathcal{E}^i(b^i; \mathbf{x})}{\partial b^i}$ as follows:

$$\frac{\partial \mathcal{E}^i(b^i; \mathbf{x})}{\partial b^i} = \beta \sum_z \pi^i(z) \left(\begin{aligned} & -\zeta^i \mathbb{I}[\rho_1^i(z) > \zeta^i b^i] \\ & + \int_{\hat{s}^i(b^i, z; \mathbf{x})}^{\bar{s}} \left(\rho_2^i(s, z) \frac{\partial \theta^i(b^i, z; \mathbf{x})}{\partial b^i} - (1 - \zeta^i) + \frac{\partial t^i(b^i, s, z)}{\partial b^i} \right) dF^i(s|z) \end{aligned} \right),$$

where the boundary term of the Leibniz rule is zero due to the optimality of the default decision. Similarly for $\frac{\partial Q^i(b^i; \mathbf{x})}{\partial b^i}$, we find that

$$\begin{aligned} \frac{\partial Q^i(b^i; \mathbf{x})}{\partial b^i} = & \beta^C \left[\zeta^i + \sum_z \pi^C(z) \left(\int_{\underline{s}}^{\hat{s}^i(b^i, z; \mathbf{x})} \left(\phi \rho_2^i(s, z) \frac{\partial \theta^i(b^i, z; \mathbf{x})}{\partial b^i} + \frac{\partial t^i(b^i, s, z)}{\partial b^i} \right) dF^{i,C}(s|z) \right. \right. \\ & + \left((\phi - 1) \rho_2^i(\hat{s}^i(b^i, z; \mathbf{x}), z) \theta^i(b^i, z; \mathbf{x}) \right) f^{i,C}(\hat{s}^i(b^i, z; \mathbf{x})|z) \frac{\partial \hat{s}^i(b^i, z; \mathbf{x})}{\partial b^i} \\ & \left. \left. + (1 - \zeta^i) [1 - F^{i,C}(\hat{s}^i(b^i, z; \mathbf{x})|z)] \right) \right], \end{aligned}$$

where $\frac{\partial t^i(b^i, s, z)}{\partial b^i} = \alpha_b^i(z)$. Note that these derivations imply that leverage is driven by i) differences in discount factors, ii) differences in beliefs, iii) default deadweight losses, and iv) the form of the bailout.

E.1.7 Equilibrium Price (Date 0)

Outsiders' date 0 optimality condition is given by $p_0 = \Upsilon'(k_0^O)$. When combined with market clearing, the date 0 equilibrium price p_0^* simply solves the following fixed point:

$$p_0^* = \Upsilon' \left(\sum_i k_0^i(p_0^*) \right),$$

where $k_0^i(p_0^*)$ is determined as in Equation (OA32).

E.1.8 Model Computation

Given our functional form assumptions, and the preceding derivations, for given regulations, computing a model solution boils down to solving a nonlinear system of 5 equations, given by

$$\begin{aligned}\frac{\partial M^R(b^R; \mathbf{x})}{\partial b^R} &= \tau_b^R \\ \frac{\partial M^U(b^U; \mathbf{x})}{\partial b^U} &= \tau_b^U \\ k_0^R &= \left(\frac{1}{\psi^R} \left(M^R(b^R; \mathbf{x}) - p_0 - \tau_k^R \right) \right)^{\frac{1}{\nu^R-1}} \\ k_0^U &= \left(\frac{1}{\psi^U} \left(M^U(b^U; \mathbf{x}) - p_0 - \tau_k^U \right) \right)^{\frac{1}{\nu^U-1}} \\ p_1(z=1; \mathbf{x}) &= \left(\zeta^R b^R k_0^R + \zeta^U b^U k_0^U + p_1(z=1; \mathbf{x}) \right)^{\frac{\gamma-1}{\gamma}},\end{aligned}$$

and 5 unknowns: b^R , b^U , k_0^R , k_0^U , and $p_1(z=1)$.

E.2 Normative Results: Detailed Derivations

E.2.1 Welfare Effects

We first characterize the marginal welfare impact on banks, creditors, and outsiders of a change in the regulations $\boldsymbol{\tau} \in \{\{\tau_b^i\}_i, \{\hat{\tau}_k^i\}_i\}$.

Banks. Leaving aside endowments, the indirect utility of banks — from the perspective of a planner with beliefs indexed by P — can be expressed as

$$V^{i,P} = \max_{b^i, k_0^i} \left\{ \left(\mathcal{E}^{i,P}(b^i; \mathbf{x}) + Q^i(b^i; \mathbf{x}) - p_0 \right) k_0^i - \Psi^i(k_0^i) - \tau_b^i b^i k_0^i - \hat{\tau}_k^i k_0^i + T_0^i \right\},$$

where $\mathcal{E}^{i,P}(b^i; \mathbf{x})$ is defined as in (OA29), but taking expectations under the planner's beliefs. Note that $Q^i(b^i; \mathbf{x})$ is the credit surface determined in equilibrium, by the creditors' beliefs, not the planner's beliefs. The marginal welfare effect of a change in regulation is thus given by

$$\begin{aligned}\frac{dV^{i,P}}{d\tau} &= \left(\frac{d\mathcal{E}^{i,P}(b^i; \mathbf{x})}{d\tau} + \frac{dQ^i(b^i; \mathbf{x})}{d\tau} - \frac{dp_0}{d\tau} \right) k_0^i + \left(\mathcal{E}^{i,P}(b^i; \mathbf{x}) + Q^i(b^i; \mathbf{x}) - p_0 - \Psi^{i'}(k_0^i) \right) \frac{dk_0^i}{d\tau} \\ &\quad - \left(\tau_b^i b^i + \hat{\tau}_k^i \right) \frac{dk_0^i}{d\tau} - \tau_b^i \frac{db^i}{d\tau} k_0^i - \left(\frac{d\tau_b^i}{d\tau} b^i + \frac{d\hat{\tau}_k^i}{d\tau} \right) k_0^i + \frac{dT_0^i}{d\tau}.\end{aligned}$$

So we can express the marginal welfare effect aggregated across banks as follows:²⁸

$$\begin{aligned} \sum_{i \in \{R, U\}} \frac{dV^{i, P}}{d\tau} &= \sum_{i \in \{R, U\}} \left(\frac{d\mathcal{E}^{i, P}(b^i; \mathbf{x})}{d\tau} + \frac{dQ^i(b^i; \mathbf{x})}{d\tau} - \frac{dp_0}{d\tau} \right) k_0^i \\ &\quad + \sum_{i \in \{R, U\}} \left(\mathcal{E}^{i, P}(b^i; \mathbf{x}) + Q^i(b^i; \mathbf{x}) - p_0 - \Psi^{i'}(k_0^i) \right) \frac{dk_0^i}{d\tau}. \end{aligned}$$

Outsiders. The indirect utility of outsiders — from the perspective of a planner with beliefs indexed by P — is given by

$$V^{O, P} = \max_{k_0^O, k_1^O(z)} \left\{ p_0 k_0^O - \Upsilon(k_0^O) + \beta \sum_z \pi^P(z) \left(H(k_1^O(z)) - p_1(z; \mathbf{x}) k_1^O(z) \right) \right\}.$$

The marginal welfare effect of a change in regulation is — using optimality/envelope theorem — given by

$$\frac{dV^{O, P}}{d\tau} = \frac{dp_0}{d\tau} k_0^O - \beta \sum_z \pi^P(z) \frac{dp_1(z; \mathbf{x})}{d\tau} k_1^O(z).$$

Creditors. Leaving aside endowments, the indirect utility of creditors — from the perspective of a planner with beliefs indexed by P — can be expressed as

$$V^{C, P} = \sum_{i \in \{R, U\}} \left(Q^{i, P}(b^i; \mathbf{x}) - Q^i(b^i; \mathbf{x}) - (1 + \kappa) \mathcal{T}^i(b^i) \right) k_0^i,$$

where $Q^i(b^i; \mathbf{x})$ is defined in Equation (OA27), and where $Q^{i, P}(b^i; \mathbf{x})$ is given by

$$Q^{i, P}(b^i; \mathbf{x}) = \beta^C \left(\zeta^i b^i + \sum_z \pi^P(z) \left(\int_{\underline{s}}^{\hat{s}(b^i, z; \mathbf{x})} (\phi \rho_2^i(s, z) \theta^i(b^i, z; \mathbf{x}) + t^i(b^i, s, z)) dF^{i, C, P}(s|z) \right. \right. \\ \left. \left. + (1 - \zeta^i) b^i \int_{s^*(b^i, z; \mathbf{x})}^{\bar{s}} dF^{i, C, P}(s|z) \right) \right),$$

where we denote the net present value for creditors of the bailout transfer per unit of investment to type i banks by²⁹

$$\mathcal{T}^i(b^i) = \beta^C \sum_z \pi^P(z) \int_{\underline{s}}^{\bar{s}} t^i(b^i, s, z) dF^{i, C, P}(s|z).$$

²⁸This result uses the fact that $\sum_i T_0^i = \sum_i \tau_b^i b^i k_0^i + \sum_i \hat{\tau}_k^i k_0^i$, which implies that

$$\sum_i \frac{dT_0^i}{d\tau} = \sum_i \left[(\tau_b^i b^i + \hat{\tau}_k^i) \frac{dk_0^i}{d\tau} + \tau_b^i \frac{db^i}{d\tau} k_0^i + \left(\frac{d\tau_b^i}{d\tau} b^i + \frac{d\hat{\tau}_k^i}{d\tau} \right) k_0^i \right].$$

²⁹It is straightforward to allow for the bailout to be a function of the decisions of all banks, as in [Dávila and Walther \(2020\)](#). In that case, $\mathcal{T}^i(b^i)$ also depends on $\mathbf{x}$, introducing additional marginal distortions.

The marginal welfare effect of a change in regulation is thus given by

$$\begin{aligned} \frac{dV^{C,P}}{d\tau} &= \sum_{i \in \{R,U\}} \left(\frac{dQ^{i,P}(b^i; \mathbf{x})}{d\tau} - \frac{dQ^i(b^i; \mathbf{x})}{d\tau} - (1 + \kappa) \frac{d\mathcal{T}^i(b^i)}{d\tau} \right) k_0^i \\ &\quad + \sum_{i \in \{R,U\}} \left(Q^{i,P}(b^i; \mathbf{x}) - Q^i(b^i; \mathbf{x}) - (1 + \kappa) \mathcal{T}^i(b^i) \right) \frac{dk_0^i}{d\tau}. \end{aligned}$$

E.2.2 Aggregate Welfare Effects

After aggregating, we can express $\frac{dW}{d\tau}$ as

$$\begin{aligned} \frac{dW}{d\tau} &= \sum_{i \in \{R,U\}} \left(\frac{dM^{i,P}(b^i; \mathbf{x})}{d\tau} - (1 + \kappa) \frac{d\mathcal{T}^i(b^i)}{d\tau} \right) k_0^i - \beta \sum_z \pi^P(z) \frac{dp_1(z; \mathbf{x})}{d\tau} k_1^O(z) \\ &\quad + \sum_{i \in \{R,U\}} \left(M^{i,P}(b^i; \mathbf{x}) - p_0 - \Psi^{i'}(k_0^i) - (1 + \kappa) \mathcal{T}^i(b^i) \right) \frac{dk_0^i}{d\tau}, \end{aligned}$$

where $M^{i,P}(b^i; \mathbf{x}) = \mathcal{E}^{i,P}(b^i; \mathbf{x}) + Q^{i,P}(b^i; \mathbf{x})$ and $\frac{dM^{i,P}(b^i; \mathbf{x})}{d\tau} = \frac{d\mathcal{E}^{i,P}(b^i; \mathbf{x})}{d\tau} + \frac{dQ^{i,P}(b^i; \mathbf{x})}{d\tau}$. Note that we can express $\frac{d\mathcal{T}^i(b^i)}{d\tau}$ and $\frac{dM^{i,P}(b^i; \mathbf{x})}{d\tau}$ as follows:

$$\begin{aligned} \frac{d\mathcal{T}^i(b^i)}{d\tau} &= \frac{\partial \mathcal{T}^i(b^i)}{\partial b^i} \frac{db^i}{d\tau} \\ \frac{dM^{i,P}(b^i; \mathbf{x})}{d\tau} &= \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial b^i} \frac{db^i}{d\tau} + \sum_z \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} \frac{dp_1(z; \mathbf{x})}{d\tau}, \end{aligned}$$

so we can express $\frac{dW}{d\tau}$ — also exploiting Equations (OA30) and (OA31) — as

$$\begin{aligned} \frac{dW}{d\tau} &= \sum_{i \in \{R,U\}} \left(\tau_b^i - \left(\frac{\partial M^i(b^i; \mathbf{x})}{\partial b^i} - \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial b^i} \right) - (1 + \kappa) \frac{\partial \mathcal{T}^i(b^i)}{\partial b^i} \right) k_0^i \frac{db^i}{d\tau} \\ &\quad + \sum_{i \in \{R,U\}} \left(\tau_k^i - \left(M^i(b^i; \mathbf{x}) - M^{i,P}(b^i; \mathbf{x}) \right) - (1 + \kappa) \mathcal{T}^i(b^i) \right) \frac{dk_0^i}{d\tau} \\ &\quad + \sum_z \left(\sum_{i \in \{R,U\}} \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} k_0^i - \beta \pi^P(z) k_1^O(z) \right) \frac{dp_1(z; \mathbf{x})}{d\tau}, \end{aligned}$$

where $\tau_k^i = \hat{\tau}_k^i + \tau_b^i b^i$ and where we can write

$$\frac{dp_1(z; \mathbf{x})}{d\tau} = \sum_i \frac{\partial p_1^*(z; \mathbf{x})}{\partial k_0^i} \frac{dk_0^i}{d\tau} + \sum_i \frac{\partial p_1^*(z; \mathbf{x})}{\partial b^i} \frac{db^i}{d\tau},$$

where $\frac{\partial p_1^*(z; \mathbf{x})}{\partial k_0^i}$ and $\frac{\partial p_1^*(z; \mathbf{x})}{\partial b^i}$ can be explicitly computed from Equation (OA25).

Hence, $\frac{dW}{d\tau}$ can be expressed as

$$\begin{aligned} \frac{dW}{d\tau} = & \sum_{i \in \{R, U\}} \left(\tau_b^i - \left(\frac{\partial M^i(b^i; \mathbf{x})}{\partial b^i} - \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial b^i} \right) - (1 + \kappa) \frac{\partial \mathcal{T}^i(b^i)}{\partial b^i} - \sum_z \pi^P(z) \frac{\delta_p(z)}{k_0^i} \frac{\partial p_1^*(z; \mathbf{x})}{\partial b^i} \right) k_0^i \frac{db^i}{d\tau} \\ & + \sum_{i \in \{R, U\}} \left(\tau_k^i - \left(M^i(b^i; \mathbf{x}) - M^{i,P}(b^i; \mathbf{x}) \right) - (1 + \kappa) \mathcal{T}^i(b^i) - \sum_z \pi^P(z) \delta_p(z) \frac{\partial p_1^*(z; \mathbf{x})}{\partial k_0^i} \right) \frac{dk_0^i}{d\tau}, \end{aligned}$$

where $\delta_p(z)$ is defined below. In a more compact form, we can express $\frac{dW}{d\tau}$ as

$$\frac{dW}{d\tau} = \sum_i \left(\tau_b^i - \delta_b^i \right) k_0^i \frac{db^i}{d\tau} + \sum_i \left(\tau_k^i - \delta_k^i \right) \frac{dk_0^i}{d\tau},$$

where we define

$$\begin{aligned} \delta_b^i &= \underbrace{\left(\frac{\partial M^i(b^i; \mathbf{x})}{\partial b^i} - \frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial b^i} \right)}_{=\delta_{b, \text{beliefs}}} + \underbrace{(1 + \kappa) \frac{\partial \mathcal{T}^i(b^i)}{\partial b^i}}_{=\delta_{b, \text{bailout}}} + \underbrace{\sum_z \pi^P(z) \frac{\delta_p(z)}{k_0^i} \frac{\partial p_1^*(z; \mathbf{x})}{\partial b^i}}_{=\delta_{b, \text{pecuniary}}} \\ \delta_k^i &= \underbrace{\left(M^i(b^i; \mathbf{x}) - M^{i,P}(b^i; \mathbf{x}) \right)}_{=\delta_{k, \text{beliefs}}} + \underbrace{(1 + \kappa) \mathcal{T}^i(b^i)}_{=\delta_{k, \text{bailout}}} + \underbrace{\sum_z \pi^P(z) \delta_p(z) \frac{\partial p_1^*(z; \mathbf{x})}{\partial k_0^i}}_{=\delta_{k, \text{pecuniary}}}. \end{aligned}$$

To complete the characterization of marginal welfare effects, we must compute $\delta_p(z)$, and to do so $\frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)}$, which is given by

$$\frac{\partial M^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} = \frac{\partial \mathcal{E}^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} + \frac{\partial Q^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)}.$$

Using banks' optimal default decisions, we can express $\frac{\partial \mathcal{E}^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)}$ and $\frac{\partial Q^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)}$ as follows:

$$\begin{aligned} \frac{\partial \mathcal{E}^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} &= \beta \pi^P(z) \left(1 - \theta^i(b^i, z; \mathbf{x}) \right) \frac{\int_{\bar{s}^i(b^i, z; \mathbf{x})} \rho_2^i(s, z) dF^{i,P}(s|z)}{p_1^*(z)} \\ \frac{\partial Q^{i,P}(b^i; \mathbf{x})}{\partial p_1(z)} &= \beta^C \pi^P(z) \left(\begin{aligned} & \phi(1 - \theta^i(b^i, z; \mathbf{x})) \frac{\int_{\underline{s}^i(b^i, z; \mathbf{x})} \rho_2^i(s, z) dF^{i,C,P}(s|z)}{p_1^*(z)} \\ & + (\phi - 1) \rho_2^i(\hat{s}^i(b^i, z; \mathbf{x}), z) \theta^i(b^i, z; \mathbf{x}) f^{i,C,P}(\hat{s}^i(b^i, z; \mathbf{x})|z) \frac{\partial \hat{s}^i(b^i, z; \mathbf{x})}{\partial p_1(z)} \end{aligned} \right), \end{aligned}$$

where we use the fact that $\frac{\partial \theta^i(\cdot)}{\partial p_1^*(z)} = \frac{1 - \theta^i(\cdot)}{p_1^*(z)}$ and where $1 - \theta^i(b^i, z; \mathbf{x}) = \frac{k_0^i - k_1^{i*}(z)}{k_0^i}$. Hence, we can

write $\delta_p(z)$ as

$$\begin{aligned} \delta_p(z) = & \sum_{i \in \{R, U\}} \left(\underbrace{\beta \frac{\int_{\hat{s}^i(b^i, z; \mathbf{x})} \rho_2^i(s, z) dF^{i, P}(s|z)}{p_1^*(z)} + \beta^C \frac{\phi \int_{\underline{s}}^{\hat{s}^i(b^i, z; \mathbf{x})} \rho_2^i(s, z) dF^{i, C, P}(s|z)}{p_1^*(z)}}_{=\lambda_1^i(z)} - \beta \right) (k_1^{i*}(z) - k_0^i) \\ & + \beta^C (1 - \phi) \sum_{i \in \{R, U\}} \rho_2^i(\hat{s}^i(b^i, z; \mathbf{x}), z) k_1^{i*}(z) \underbrace{f^{i, C, P}(\hat{s}^i(b^i, z; \mathbf{x})|z)}_{=\frac{\partial \mathcal{F}^i(z)}{\partial p_1(z)}} \frac{\partial \hat{s}^i(b^i, z; \mathbf{x})}{\partial p_1(z)}. \end{aligned}$$

E.3 Scenarios

Here we provide explicit formulae for the different regulatory scenarios and experiments considered in Section 4.2. Note that the first-best scenario is straightforward, and simply characterized by $\tau_b^i = \delta_b^i$ and $\tau_k^i = \delta_k^i$, as implied by Lemma 1.

E.3.1 Unconstrained-Leverage-Regulation Scenario

In the unconstrained-leverage-regulation scenario, the optimal regulation satisfies

$$\begin{aligned} \frac{dW}{d\tau_b^R} &= (\tau_b^R - \delta_b^R) \frac{db^R}{d\tau_b^R} k_0^R + (\tau_b^U - \delta_b^U) \frac{db^U}{d\tau_b^R} k_0^U - \delta_k^R \frac{dk_0^R}{d\tau_b^R} - \delta_k^U \frac{dk_0^U}{d\tau_b^R} = 0 \\ \frac{dW}{d\tau_b^U} &= (\tau_b^R - \delta_b^R) \frac{db^R}{d\tau_b^U} k_0^R + (\tau_b^U - \delta_b^U) \frac{db^U}{d\tau_b^U} k_0^U - \delta_k^R \frac{dk_0^R}{d\tau_b^U} - \delta_k^U \frac{dk_0^U}{d\tau_b^U} = 0, \end{aligned}$$

or solving for the optimal leverage regulations $\tau_b^R - \delta_b^R$ and $\tau_b^U - \delta_b^U$ in matrix form,

$$\begin{pmatrix} \tau_b^R \\ \tau_b^U \end{pmatrix} = \begin{pmatrix} \delta_b^R \\ \delta_b^U \end{pmatrix} + \begin{pmatrix} \frac{db^R}{d\tau_b^R} k_0^R & \frac{db^U}{d\tau_b^R} k_0^U \\ \frac{db^R}{d\tau_b^U} k_0^R & \frac{db^U}{d\tau_b^U} k_0^U \end{pmatrix}^{-1} \begin{pmatrix} \delta_k^R \frac{dk_0^R}{d\tau_b^R} + \delta_k^U \frac{dk_0^U}{d\tau_b^R} \\ \delta_k^R \frac{dk_0^R}{d\tau_b^U} + \delta_k^U \frac{dk_0^U}{d\tau_b^U} \end{pmatrix},$$

as implied by Proposition 1.

E.3.2 Constrained-Leverage-Regulation Scenario

In the constrained-leverage-regulation scenario, the optimal leverage regulation of regulated banks satisfies

$$\frac{dW}{d\tau_b^R} = (\tau_b^R - \delta_b^R) \frac{db^R}{d\tau_b^R} k_0^R - \delta_b^U \frac{db^U}{d\tau_b^R} k_0^U - \delta_k^U \frac{dk_0^U}{d\tau_b^R} - \delta_k^R \frac{dk_0^R}{d\tau_b^R} = 0,$$

or solving for τ_b^R ,

$$\tau_b^R = \delta_b^R - \left(-\frac{db^R}{d\tau_b^R} \right)^{-1} \left(\delta_k^U \frac{d \log k_0^U}{d\tau_b^R} \frac{k_0^U}{k_0^R} + \delta_k^R \frac{d \log k_0^R}{d\tau_b^R} + \delta_b^U \frac{db^U}{d\tau_b^R} \frac{k_0^U}{k_0^R} \right).$$

This expression highlights the role played by the three estimated leakage elasticities in Table 1: $\frac{d \log k_0^U}{d\tau_b^R} / \left(-\frac{db^R}{d\tau_b^R} \right)$, $\frac{d \log k_0^R}{d\tau_b^R} / \left(-\frac{db^R}{d\tau_b^R} \right)$, and $\frac{db^U}{d\tau_b^R} / \left(-\frac{db^R}{d\tau_b^R} \right)$ — Footnote 12 explains the normalization by $\frac{db^R}{d\tau_b^R}$ that we use to compute tax elasticities. In this case, the marginal welfare change associated with changing the leverage regulation on shadow banks in the constrained-leverage-regulation scenario is given by

$$\frac{dW}{d\tau_b^U} = \left(\tau_b^U - \delta_b^U \right) \frac{db^U}{d\tau_b^U} k_0^U - \delta_k^U \frac{dk_0^U}{d\tau_b^U} - \delta_k^R \frac{dk_0^R}{d\tau_b^U} + \left(\tau_b^R - \delta_b^R \right) \frac{db^R}{d\tau_b^U} k_0^R.$$

Note that we can express $\frac{dW}{d\tau_b^U}$ using the elements of the Le Chatelier matrix as follows:

$$\frac{dW}{d\tau_b^U} = \left(\tau_b^U - \delta_b^U \right) \frac{db^U}{d\tau_b^U} k_0^U \left(1 - L_b^U \right) + \left(-\delta_k^U \right) \frac{dk_0^U}{d\tau_b^U} \left(1 - L_k^U \right) + \left(-\delta_k^R \right) \frac{dk_0^R}{d\tau_b^U} \left(1 - L_k^R \right),$$

where

$$L_b^U = \left(\frac{db^U}{d\tau_b^U} \right)^{-1} \frac{db^R}{d\tau_b^U} \left(\frac{db^R}{d\tau_b^R} \right)^{-1} \frac{db^U}{d\tau_b^R}, \quad L_k^U = \left(\frac{dk_0^U}{d\tau_b^U} \right)^{-1} \frac{db^R}{d\tau_b^U} \left(\frac{db^R}{d\tau_b^R} \right)^{-1} \frac{dk_0^U}{d\tau_b^R},$$

and

$$L_k^R = \left(\frac{dk_0^R}{d\tau_b^R} \right)^{-1} \frac{db^R}{d\tau_b^R} \left(\frac{db^R}{d\tau_b^R} \right)^{-1} \frac{dk_0^R}{d\tau_b^R}.$$

E.3.3 Uniform-Leverage-Regulation Scenario

In the uniform-leverage-regulation scenario with unconstrained leverage, the regulator sets $\tau_b^U = \tau_b^R = \bar{\tau}_b$ to maximize $\frac{dW}{d\bar{\tau}_b}$, given by

$$\frac{dW}{d\bar{\tau}_b} = \sum_i \left(\bar{\tau}_b - \delta_b^i \right) \frac{db^i}{d\bar{\tau}_b} k_0^i + \sum_i \left(-\delta_k^i \right) \frac{dk_0^i}{d\bar{\tau}_b},$$

which yields the formula for the optimal uniform leverage regulation

$$\bar{\tau}_b = \frac{\sum_i \frac{db^i}{d\bar{\tau}_b} k_0^i \delta_b^i + \sum_i \frac{dk_0^i}{d\bar{\tau}_b} \delta_k^i}{\sum_i \frac{db^i}{d\bar{\tau}_b} k_0^i}.$$